Step Tracker System

SYSTEM DATASHEET Rev. A

Features

- Arduino UNO R4 WiFi
- GC9A01 1.28” round 240×240 LCD display
- ADXL335 3-axis analog accelerometer
- Sallen-Key Low Pass Filter Circuit
- 3-step boot calibration sequence with LCD UI
- Adaptive step counting for Stationary / Walking / Running with added pace classification
- ADXL335 self-test via user button
- Pace Indicator lights - Red / Yellow / Green
- Digital and Analog Clocks when using WiFi
- Artificial Horizon style Attitude Indicator
- Powered using a replaceable 9V battery

Applications

- Fitness and daily activity tracking
- Accurate timekeeping device

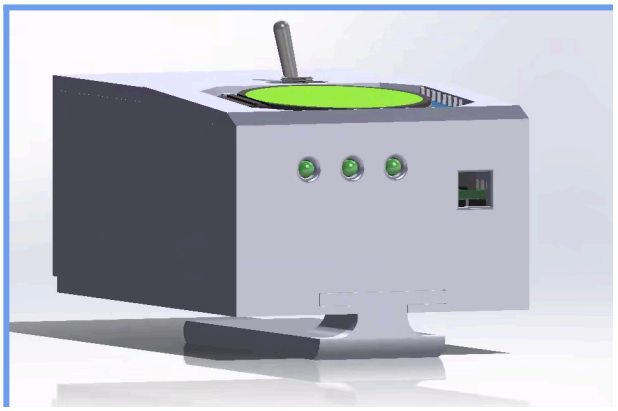
General Description

The Step Tracker is a belt-worn Arduino based system that counts user steps and identifies their pace; Stationary, Walking or Running and displays the information through an LCD screen.

It utilises an ADXL335 accelerometer paired with a Sallen-Key Low Pass Filter to pass accurate signals to the Arduino ADC, where the software will process the data and display the results on the LCD.

The system is powered by a 9V battery, and includes a guided calibration routine which allows for accurate accelerometer readings.

System Photo and 3D Render



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System Overview and Specifications

1. System and application introduction

The StepTracker v1.0 is a high precision, small form factor step tracking system designed to be a wearable device for portable data tracking. The system utilises a dual board architecture, combining an Arduino Uno R4 microcontroller with a custom PCB shield allowing for the functionality of various subsystems. It is designed to be powered through a 9v battery or a lab power supply. The onboard ADXL335 accelerometer captures acceleration in 3 axes, allowing for accurate step tracking. Added Wifi capability also allows for accurate analog and digital clocks.

2. Highlighted system features

2.1 Power supply

- 9 volt battery power
- External power barrel input
- 3-terminal switch between battery power and external barrel jack input
- Overcurrent / overvoltage / reverse-polarity protection
- Two regulators to step down from 9V to 5V (Arduino), 5V to 3.3V (Accelerometer)
- Indicator LEDs per power rail

2.2 Accelerometer

- Chipset model: ADXL335
- Input voltage 1.8V - 3.6V (3.3V preferred)
- X, Y, Z axis

2.3 Low-Pass Filter

- Noise filtering of X, Y, and Z
- Clean signals for easy steps calculation

2.4 System Peripherals

- Arduino UNO Controlled
- GC9A01A LCD Display
- 3x Indicator LED’s for walk pace identification each LED (red - stationary, yellow-walking, green - running)

3. Functional and hardware requirements list

Functional/Software Requirements

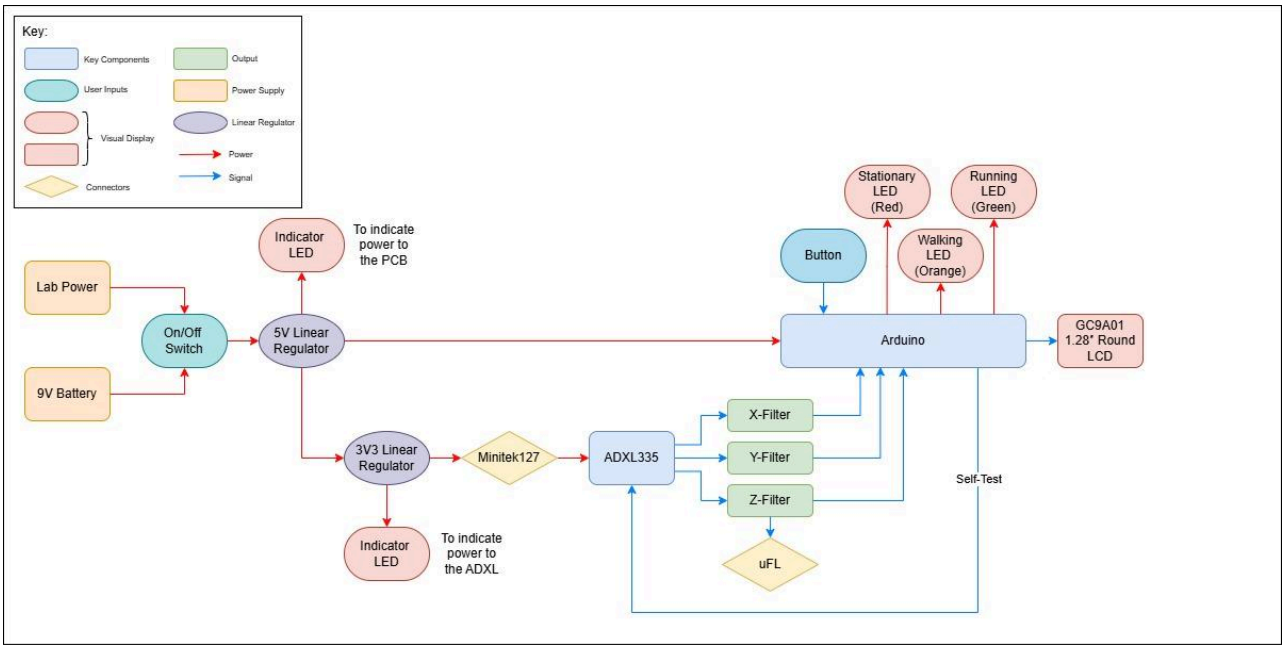
#	Software	Requirement	Verification
1	**Self-Test Routine and Feedback**	The system shall perform an ADXL335 self-test using the ST pin, verify measured responses against datasheet limits, and display a PASS/FAIL result on the LCD within 5 seconds.	Compare measured ΔX, ΔY, ΔZ values against datasheet limits.
2	**Static Acceleration Calibration**	The system shall calibrate accelerometer offsets in six orientations (+X, -X, +Y, -Y, +Z, -Z) and store calibration values in EEPROM for retention after power cycling.	Compare outputs against expected gravity values and perform a power-cycle test.
3	**Step Counting**	The system shall count steps using filtered accelerometer data, provide a push-button reset function, and maintain a step count error of less than ±10% during normal walking.	Compare recorded steps to manual counts during a controlled walking test.
4	**Walking Pace Identification**	The system shall classify motion as stationary, walking, or running and indicate the detected pace using dedicated LEDs.	Walking and running trials.
5	**LCD Display**	The LCD shall display step count, pace status, calibration status, and self-test results.	Visual inspection.
6	**ADC Sampling**	The Arduino ADC shall sample all three filtered accelerometer outputs at a minimum rate of 100 Hz.	Firmware verification.
7	**User Input Handling**	Push button inputs shall be debounced in software with a debounce period of at least 20 ms.	Firmware verification.
8	**System Startup**	The system shall initialise and become operational within 3 seconds of power-up.	Timing test.

Hardware Requirements

#	Hardware	Requirement	Verification
1	Standards Compliance	Conform with AS/NZS 3000 and all UTS guidelines.	Design review.
2	Enclosure and Wearability	Wearable design with total mass less than 300 g and enclosure dimensions not exceeding the size of the largest component.	Measurement.
3	Custom PCB shield	Include test points for 5 V, 3.3 V, GND, and filter inputs/outputs.	PCB inspection.
3.1	Custom PCB shield	Must fit onto Arduino Uno R4 (68.6 mm × 53.3 mm).	Mechanical fit test.
4	PCB Power Supply	Provide regulated 5 V and 3.3 V supply rails and a power indicator LED.	Voltage measurement and functional test.
4.1	PCB Power Supply	Supply the ADXL335 with 3.3 V	Voltage measurement.
4.2	PCB Power Supply	Include a user-accessible on/off switch.	Functional test.
4.3	PCB Power Supply	Support operation from both barrel jack and battery power (+9V)	Functional and battery test.
5	Amphenol ICC Minitek127 connector	Interface with the ADXL335 breakout PCB using the correct pin configuration.	Continuity and functional test.
5.1	Amphenol ICC Minitek127 connector	Implement circuitry required for the ADXL335 self-test function.	Self-test verification.
6	Signal Conditioning (Analog Filtering)	Implement second-order low-pass filtering on all three accelerometer axes with a cutoff frequency of 100 Hz ±10%.	Frequency response measurement.
7	Interface between the the PCB shield and Arduino and measurement equipment	Connect filtered accelerometer outputs to Arduino ADC inputs for digitisation.	PCB inspection and functional test.
7.1	8mm mating length headers	Connect the PCB shield to the Arduino Uno R4.	Mechanical inspection.

#	Hardware	Requirement	Verification
7.2	UFL → BNC Connector	Provide access to the filtered Z-axis signal for oscilloscope measurement.	Oscilloscope test.
8	Power Switch	Mounted on the PCB shield and accessible through the enclosure.	Visual inspection.
8.1	User input button	Provide calibration, mode selection, and step count reset functions.	Functional test.
9	Visual Outputs	Each power supply rail shall have an indicator LED.	Functional test.
9.1	Visual Outputs	System power LED shall be visible when the PCB is mounted in the enclosure.	Visual inspection.
9.2	Visual Outputs	Dedicated LEDs shall indicate stationary, walking, and running states.	Functional test.
9.3	Visual Outputs	LCD shall display system status and step count information.	Functional test.
10	Mechanical integration	PCB shall fit within the enclosure and provide access to LEDs, buttons, LCD, connectors, and mounting hardware.	Assembly and fit test.

4. System block diagram



The diagram above presents the overall system block diagram for the step tracker, illustrating the signal and power flow between all major subsystems. The system is powered by either a 9V battery or laboratory power supply, selected via an on/off switch and regulated to 5V and 3.3V by two linear regulators. The 5V rail supplies the Arduino and peripherals, and the 3.3V rail supplies the ADXL335 accelerometer via the Minitek127 connector. The ADXL335 outputs three analogue acceleration signals along the X, Y, and Z axes, each passed through a dedicated filter before being read by the Arduino. A uFL connector is provided on the ADXL335 for filter response verification. The Arduino processes the filtered signals and drives three activity indicator LEDs, stationary (red), walking (orange), and running (green) as well as a round LCD display. A pushbutton provides user input. Indicator LEDs on each power rail confirm that the PCB and accelerometer are correctly powered.

5. Signal Mapping Table

CUSTOM PCB - STEPTRACKER(1.0)

Signal Name	Source	Destination	Source Pin	Dest.. Pin	Type	Description
9V	b2b-ph-k-s(lf)(sn) (Terminal pins for voltage source)	ANT23SECQE (3-terminal switch)	1	1_NC1	DC Analog	9 volts of DC power from a 9 volt battery or power barrel will first enter into the first regulator for the voltage to be stepped down to 5 volts.
	PJ-102AH (Power jack) **Alternate source**		3	3_NO1		
9V	ANT23SECQE (3-terminal switch)	LM7805CT (Linear voltage regular)	5_COM2	1_IN	DC Analog	The 3-terminal switch depending on the switch's position will determine which source will the power source be. One side toggled, will come from the 9 volt battery and the other side toggle will be for 9 volt power jack
5V	LM7805CT (Linear voltage regular)	LD1117S33TR (Low dropout regulator)	3_OUT	3_VIN	DC Analog	5 volts of DC power from the first regulator will be further stepped down to 3.3 volts.
				Pins requiring 5 volts		
3V3	LD1117S33TR (Low dropout regulator)	Amphenol ICC Minitek127 ADXL335 (Accelerometer)	2_OUT	5	DC Analog	3.3 volts from the second regulator will be used to power the ADXL335 accelerometer.
X_OUT	Amphenol ICC Minitek127 ADXL335 (Accelerometer)	MCP6004-I/SL (Ampliflier 1)	2_2	3_+	Unfiltered XYZ Analog Signal	X,Y, and Z signals will be translated from the acceleration of x, y, and z coordinates, however it becomes an unfiltered signal.
Y_OUT		MCP6004-I/SL (Ampliflier 2)	4_4	5_+		
Z_OUT		MCP6004-I/SL (Ampliflier 3)	6_6	12_+		
5V	LM7805CT (Linear voltage regular)	MCP6004-I/SL (Ampliflier 1)	3_OUT	4	DC Analog	5 volts to power the 3 amplifiers for low pass filter.
		MCP6004-I/SL (Ampliflier 2)	3_OUT	4		
		MCP6004-I/SL (Ampliflier 3)	3_OUT	4		
5V	LM7805CT (Linear voltage regular)	DS1 (Green LED)	3_OUT	ANODE	DC Analog	The LEDs powered from 5 volts indicated with mode of motion are the step tracker in. Green means walking, yellow means slight jogging or speed walking, and red means running.
		DS2 (Yellow LED)	3_OUT	ANODE		
		DS3 (Red LED)	3_OUT	ANODE		
5V	LM7805CT (Linear voltage regular)	BCS-104-L-S-TE (Terminal pins for LCD Display)	3_OUT	1	DC Analog	5 volts will provide the power for the LCD to be displayed.
5V	LM7805CT (Linear voltage regular)	SW1FSM2JART (Switch)	3_OUT	2	DC Analog	5 volts to power the button.

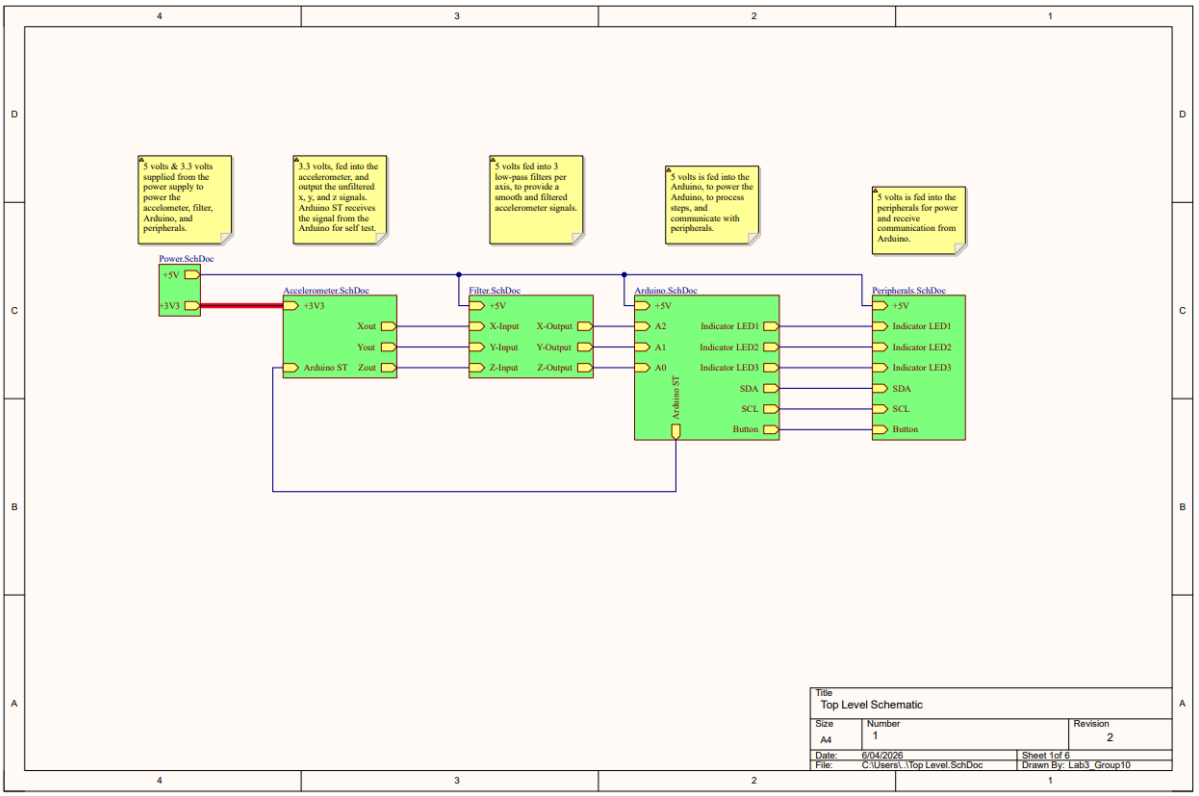
NOTE Some components such as resistors, capacitors, diodes, fuses, and some MOSFETs will not be referenced, however certain MOSFETS will be referenced for digital switching.

Arduino UNO R4

Signal Name	Source	Destination	Source Pin	Dest.. Pin	Type	Description
5V	LM7805CT (Linear voltage regular)	PREC008SAN-RC_J5 (Peripheral Terminals)	3_OUT	5	DC Analog	Power from the 5V rail into the 5V pin of the Arduino.
X_OUTPUT	MCP6004-I/SL (Ampliflier 1)	22-28-4060 (Peripheral Terminals)	1	3	Filtered XYZ Analog Signal	The filtered X,Y, and Z will be fed into the Arduino so that it can be read for processing the steps.
Y_OUTPUT	MCP6004-I/SL (Ampliflier 2)		7	2		
Z_OUTPUT	MCP6004-I/SL (Ampliflier 3)		14	1		
SDA	22-28-4060	BCS-104-L-S-TE (Terminal pins for LCD Display)	5	3	Digital Serial Data	A bi-direction line used for transferring data and inter-communication between Arduino and LCD Display.
SCL	22-28-4060	BCS-104-L-S-TE (Terminal pins for LCD Display)	6	4	Digital Clock Signals	Communicates with the LCD, when to transfer data as well and to synchronise timing for effective data transfer.
Arduino ST	PREC008SAN-RC_J6	Amphenol ICC Minitek127 ADXL335 (Accelerometer)	2	1 via PMOS 5 along with 3.3 volts	DC Digital	Arduino sends a signal to the MOSFET for self test routine input.
IND_LED1	PREC008SAN-RC_J6	2N7002H6327XTSA2 (NMOS)	3	1	DC Digital	Arduino sends a signal to the NMOSs to switch on and off the LED's.
IND_LED2		2N7002H6327XTSA2 (NMOS)	4	1		
IND_LED3		2N7002H6327XTSA2 (NMOS)	5	1		
Button	PREC008SAN-RC_J6	SW1FSM2JART (Switch)	6	1	DC Digital	Button is pressed to switch subroutines on the steptracker.

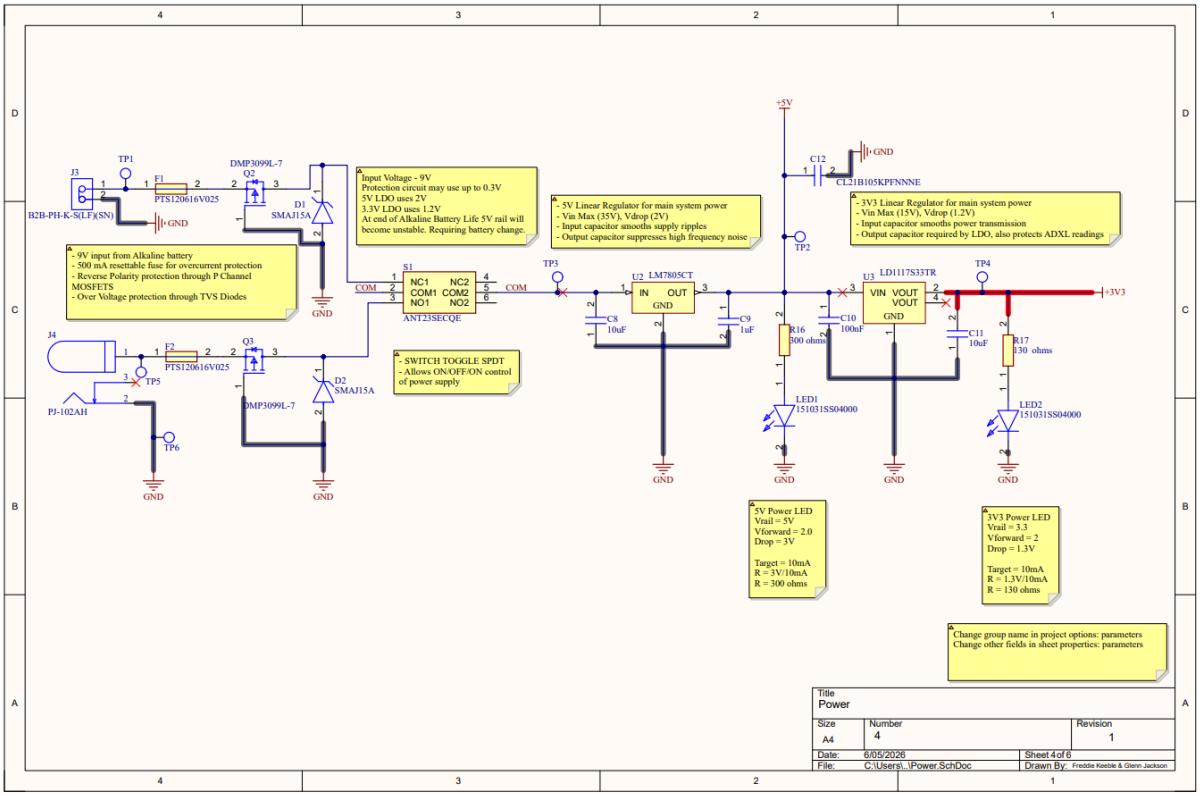
Final Schematic Design

1. Top-level



The top-level schematic implements a hierarchical design structure, interconnecting the accelerometer interface, analog filter, Arduino interface, and peripheral systems. Signals are routed between sheets using consistent net labels to improve readability and maintainability. The X, Y, and Z accelerometer outputs are routed from the accelerometer interface to the filter stage before being connected to the Arduino analog inputs. Control signals for the self-test circuit, LEDs, pushbutton, and LCD interface are also distributed through the top-level sheet.

2. Power Supply

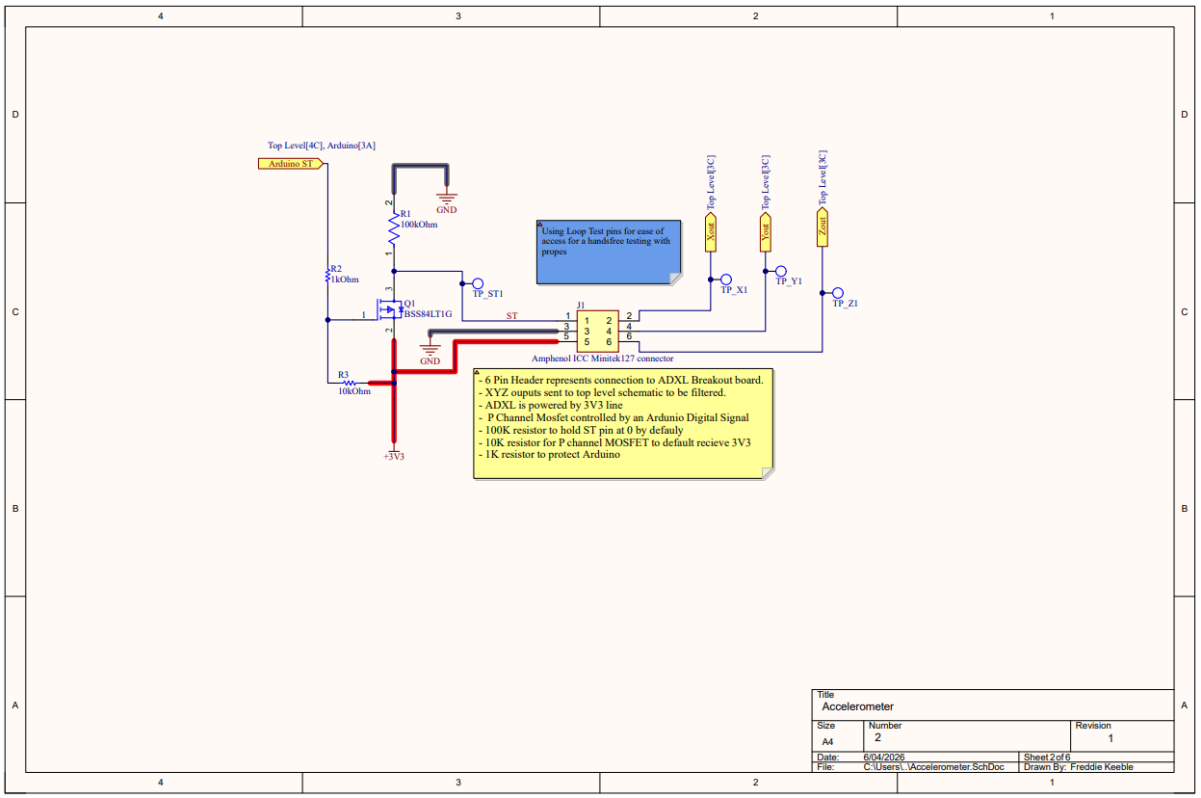


The power supply subsystem is designed to deliver stable and regulated power rails to the Step Tracker. It has a dual input architecture with individual fault protection circuitry, 2 stage voltage regulation and visual status indicator LEDs.

The system accepts power from a standard 9v battery through a JST connector, as well as a DC power barrel jack for laboratory operation. Each power source is managed via a 6 terminal Double-Pole Double-Throw (DPDT) switch that isolates the unused source. To safeguard the microcontroller and sensitive analog components, each input path utilises 3 stages of protection. A resettable PPTC fuse handles overcurrent events (>500mA) by tripping until the fault is cleared. A P-channel MOSFET is configured to allow voltage through during normal operation while blocking current if the battery or barrel jack is connected incorrectly. A TVS diode clamps high voltage spikes above the 15V threshold.

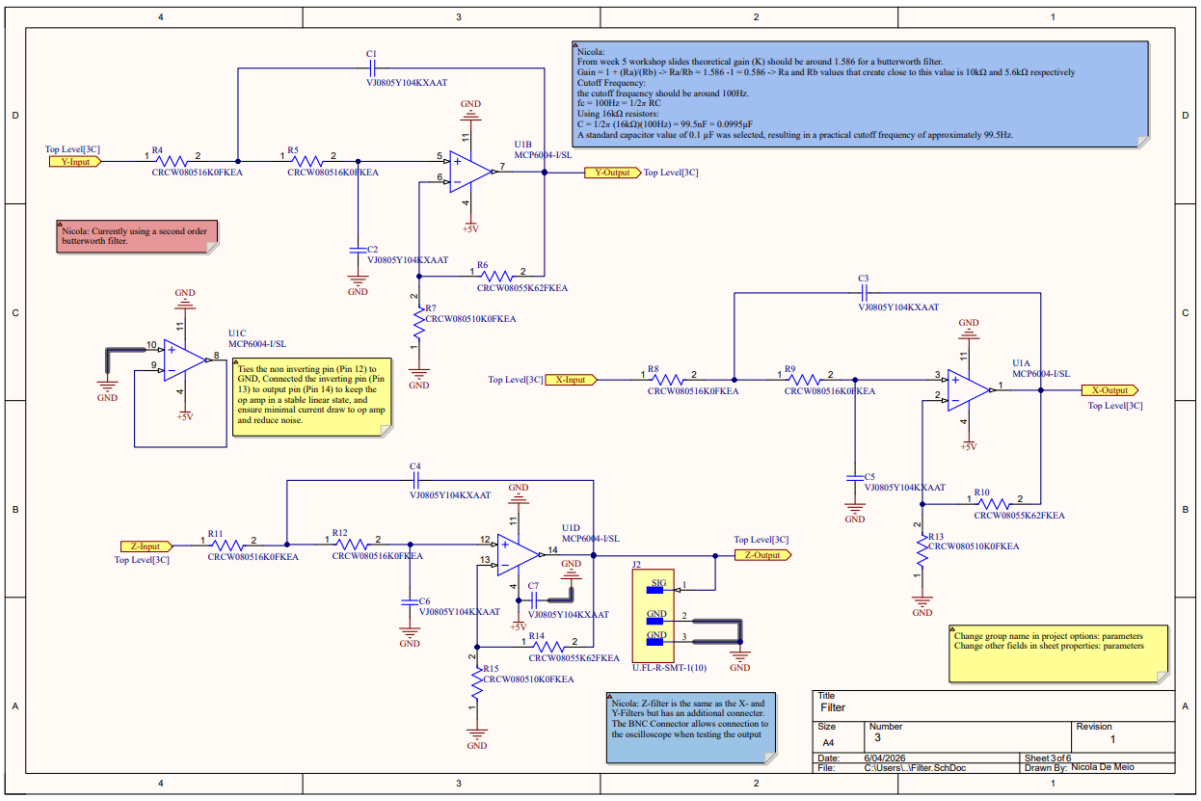
The system has both a 5V and 3.3V voltage rail, which are provided through the dual stage voltage regulation design. The 9V input is regulated down to 5V through the first LDO where it powers the Op-Amps, Arduino and system peripherals. An indicator LED tells the user if the 5V rail is operational. The stable 5V is then fed to the 3.3V LDO, which outputs a stable low noise supply for the accelerometer. This secondary rail also features an indicator LED.

3. Accelerometer Interface



The accelerometer interface provides the connection between the ADXL335 breakout board and the signal conditioning circuitry. The sensor is powered from the regulated 3.3V supply and exposes the XOUT, YOUT, and ZOUT analog outputs for subsequent filtering. A P-channel MOSFET is used to control the accelerometer's self-test (ST) input from an Arduino digital output. A 100kΩ resistor ensures the ST pin is held low during normal operation, while a 10kΩ pull-up resistor ensures the MOSFET remains off by default. A 1kΩ series resistor provides protection for the Arduino output pin. Dedicated test points are included on the self-test and axis output signals to simplify validation and troubleshooting.

4. Analog Filter



The analog filter stage consists of three identical second-order Sallen-Key Butterworth low-pass filters, one for each accelerometer axis. The filters attenuate high-frequency noise while preserving motion-related signals. A Butterworth response was selected to provide a maximally flat passband response and minimise amplitude distortion within the operating frequency range. The fourth op-amp within the MCP6004 package is configured as a unity-gain follower to prevent unused amplifier sections from floating and introducing noise into the system.

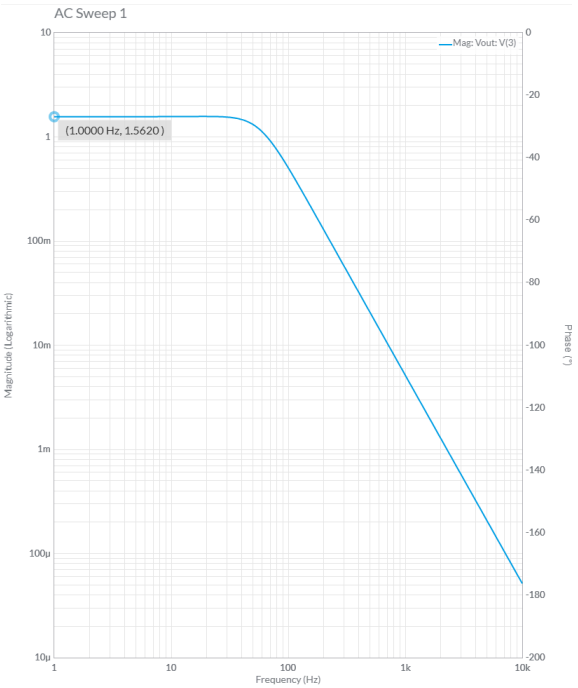
4.1 Filter Design Equations

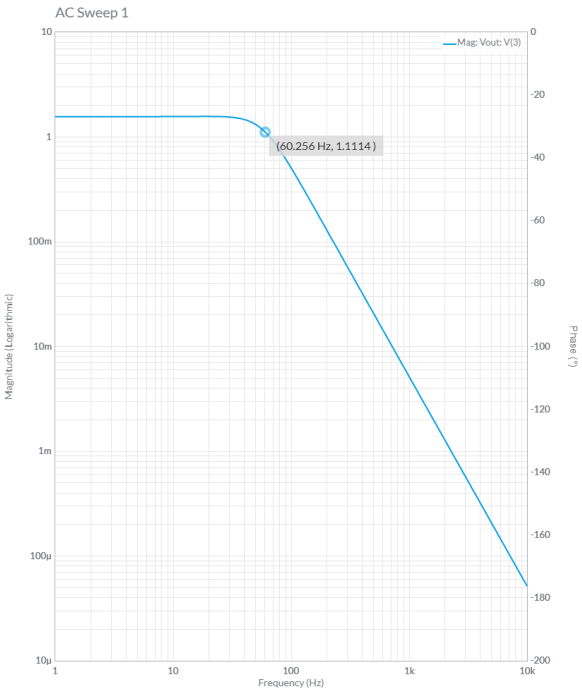
Parameter	Formula	Value	Notes
Cut-off frequency (theoretical)	$f_c = \frac{1}{2\pi RC}$	99.47 Hz	R = 16 kΩ C = 0.1μF The ADXL335 output resistance (~32 kΩ) was not included in this theoretical analysis.
Cut-off frequency (including ADXL Resistance)	$f_c = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}$	57.43 Hz	R_1 = 48 kΩ R_2 = 16 kΩ $C_1 = C_2$ = 0.1μF

Parameter	Formula	Value	Notes
Passband Gain (V/V)	$K = 1 + \frac{R_a}{R_b}$	1.56 V/V	R_a = 5.6 kΩ, R_b = 10 kΩ
-3 dB Magnitude	$\frac{K}{\sqrt{2}}$	1.104 V/V	Magnitude corresponding to the filter cutoff frequency.
Simulated Cut-off Frequency	AC Sweep Measurement	~ 60 Hz	Includes ADXL335 output resistance (~32 kΩ)

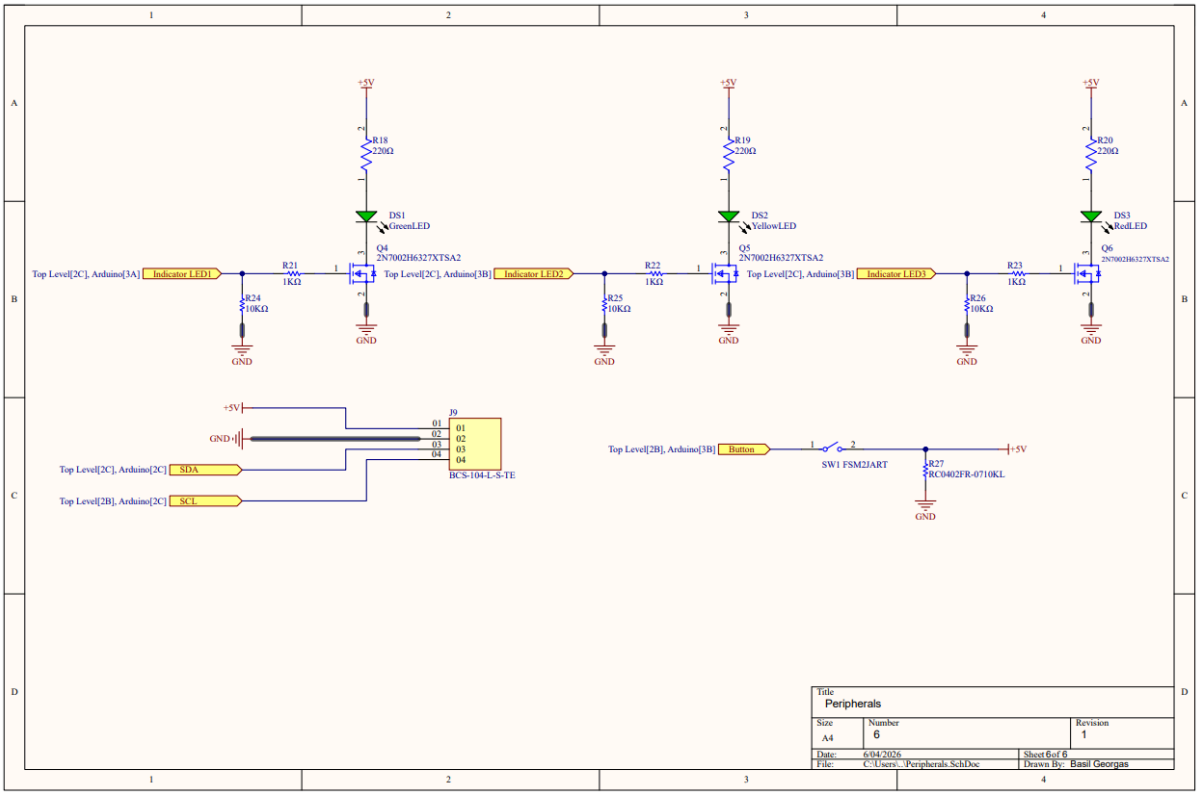
4.2 Filter Simulations

The filter response was verified using Multisim AC sweep simulations. The simulated frequency response produced a passband gain of approximately 1.56 V/V, matching the design target. The theoretical cutoff frequency of 99.47 Hz was calculated using the standard Sallen-Key approximation, assuming an ideal voltage source and therefore neglecting the ADXL335 output resistance. When the ADXL335 output resistance (approximately 32 kΩ) was included in the simulation model, the effective first-stage resistance increased to 48kΩ, yielding a theoretical cutoff frequency of approximately 57.4 Hz. This closely matches the simulated cutoff of 60 Hz, with the small remaining discrepancy attributable to op-amp input impedance and gain network loading effects not captured by the simplified formula. This reduction in cutoff frequency demonstrates the importance of considering sensor output characteristics during filter design. Identical filter sections are implemented for the X-, Y-, and Z-axis accelerometer outputs.





5. System Peripherals, System Inputs/Outputs



The peripherals subsystem provides user feedback and interaction capabilities. Three indicator LEDs are connected through 220Ω current-limiting resistors and are switched using N-channel MOSFETs, reducing current demand on the Arduino I/O pins. An external pushbutton input is provided with a 10kΩ pull-down resistor to ensure a defined logic-low state when the button is not pressed. The subsystem also includes an I2C connector providing SDA and SCL signals for interfacing with the LCD display module. The display interface was updated from the originally designed I2C LCD module to a GC9A01 1.28" round 240×240 LCD, communicating via SPI. This change is not yet reflected in the schematic but will be incorporated in future revisions.

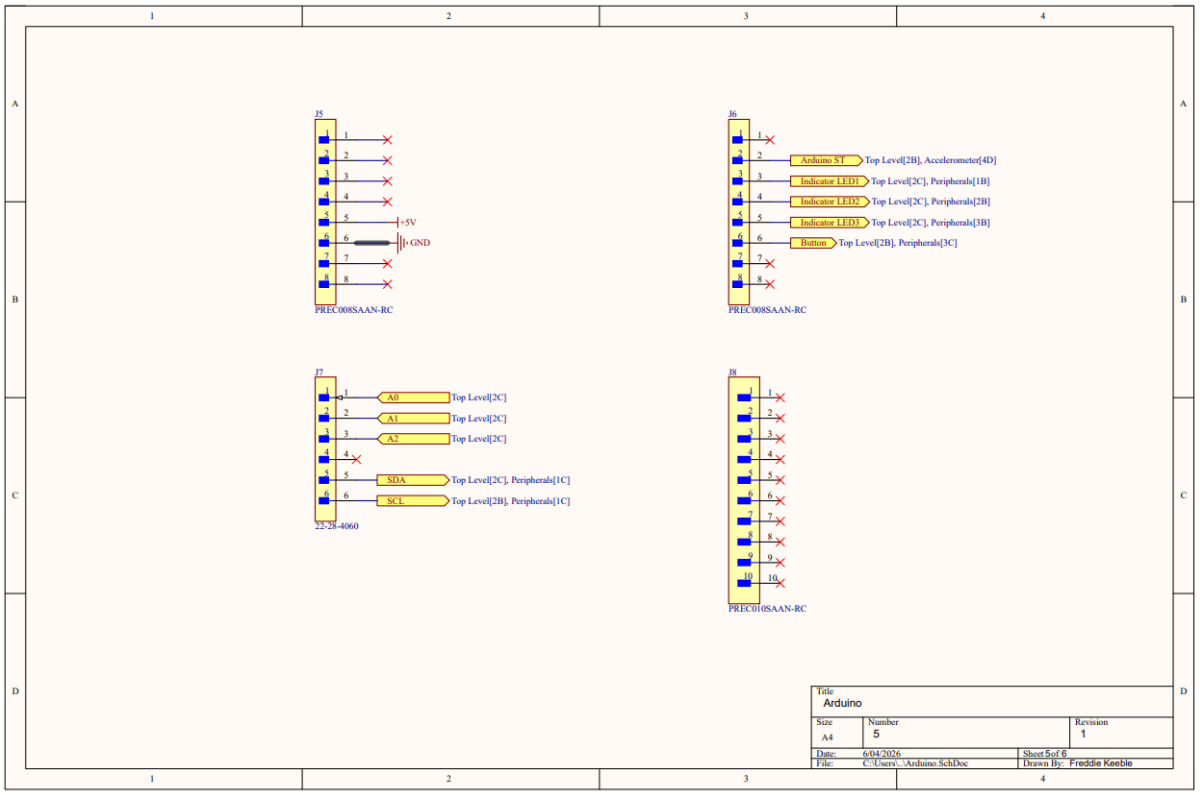
5.1 Design Equations

Parameter	Formula	Value	Notes
Green LED Resistor	$I_f = \frac{V_{supply} - V_f}{R}$	12.7 mA	V_{supply} = 5 V V_f = 2.2 V Selected R = 220Ω Calculated I_f is below the 20mA maximum
Yellow LED resistor	$I_f = \frac{V_{supply} - V_f}{R}$	13.2 mA	V_{supply} = 5 V V_f = 2.1 V Selected R = 220Ω Calculated I_f is below the 20mA maximum

Parameter	Formula	Value	Notes
Red LED resistor	$I_f = \frac{V_{supply} - V_f}{R}$	13.6 mA	V_{supply} = 5 V V_f = 2.1 V Selected R = 220Ω Calculated I_f is below the 20mA maximum

The 220Ω was chosen as a standard available resistor value, and by back-calculating the actual forward current for each LED colour we verify that none exceed the 20 mA rated maximum. The different I_f values across the three LEDs reflect the slightly different V_f of each colour.

6. Arduino Interface

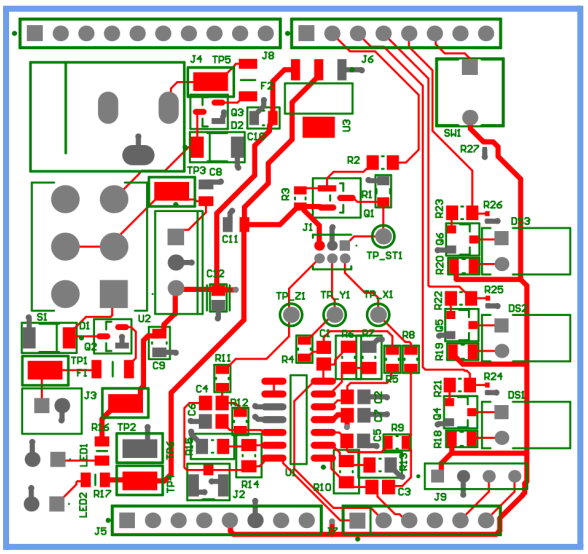


The Arduino interface sheet provides the physical connection between the Arduino UNO R4 Wifi and the rest of the system hardware. The filters X-, Y-, and Z-axis signals are connected to analog input pins A0, A1, and A2, respectively. Additional digital I/O connections are provided for the accelerometer self-test control signal, indicator LEDs, pushbutton input, and I2C communication lines used by the display module. This sheet primarily serves as a connector and signal-routing interface between the Arduino and external circuitry.

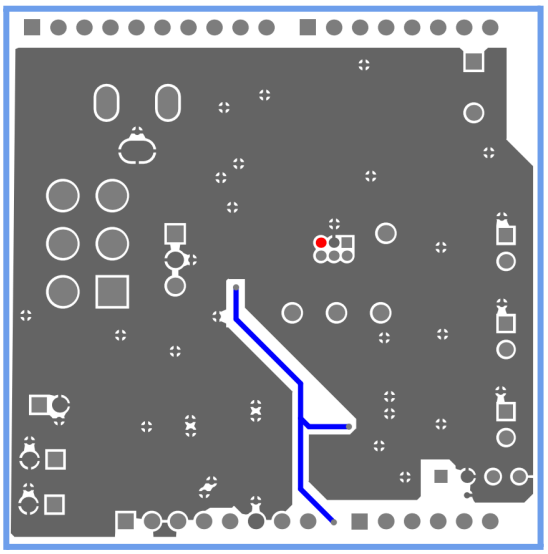
Final PCB Design (ECAD)

The StepTracker PCB was designed and engineered with adherence to high sensitivity and low noise electronics design principles. The subcircuits, component placement and orientation, routing and board shape were all optimised to ensure signal integrity, allow for simple and efficient manufacturing, and also provide the user with easy access controls in a small form factor.

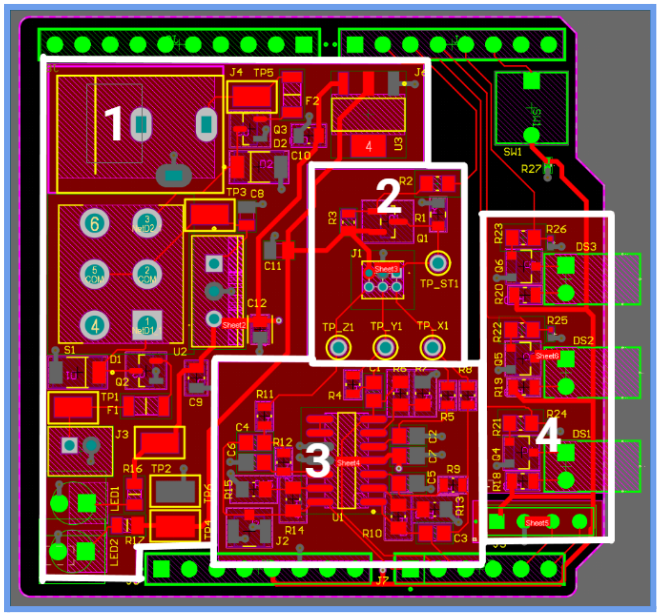
1. Physical Layout and PCB Manufacturing Design



Top Layer + Silk Screen

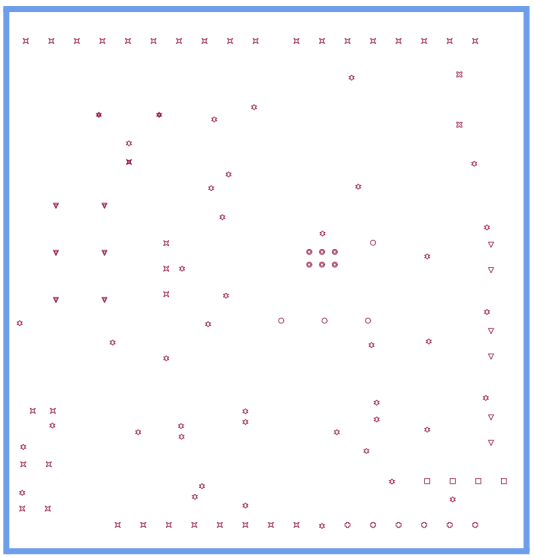


Bottom Layer



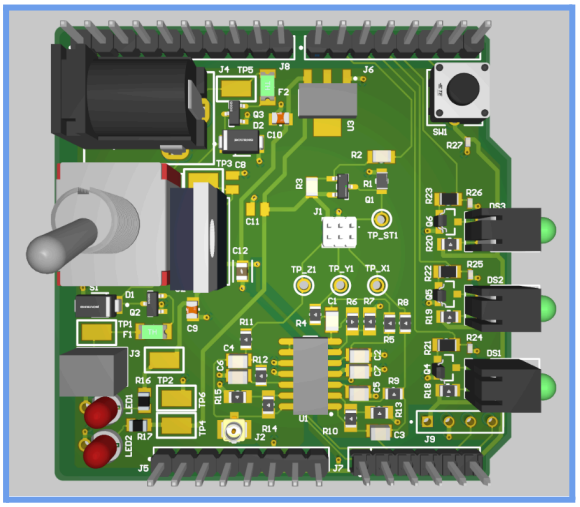
The physical layout is logically split into separate zones for each sub circuit. This allows for sensitive analog components to have small signal paths, and make debugging and maintenance efficient. The placement of the headers on the edge lines up perfectly with the female headers of the Arduino Uno R4. The top layer prioritises signal and power, with added test points, clear silkscreen and designators for every part. The bottom layer houses the ground plane and one 5V line, allowing low impedance return path for all subcircuits. The physical boundaries of the PCB mirror the outline of the Arduino Uno R4.

Board Outline

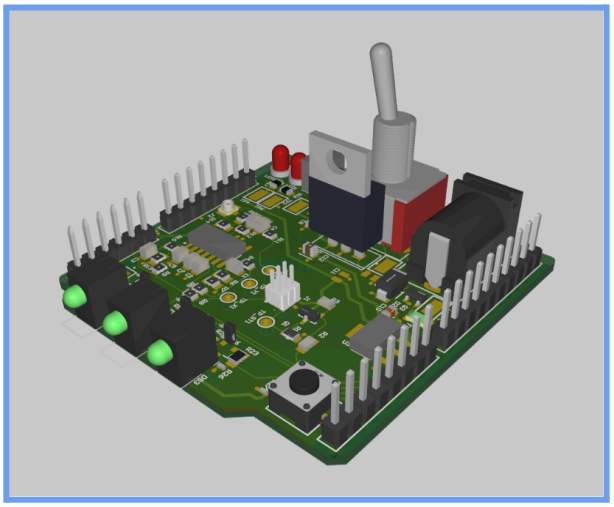


Drill Drawing

2. PCB 3D Print

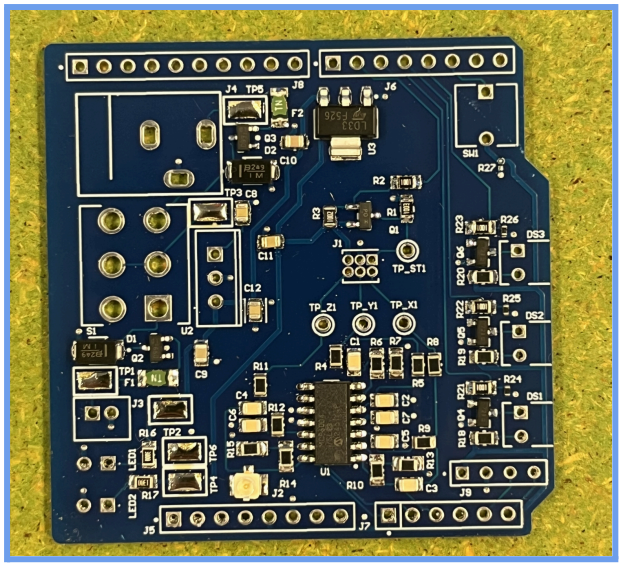


PCB Render - Top View

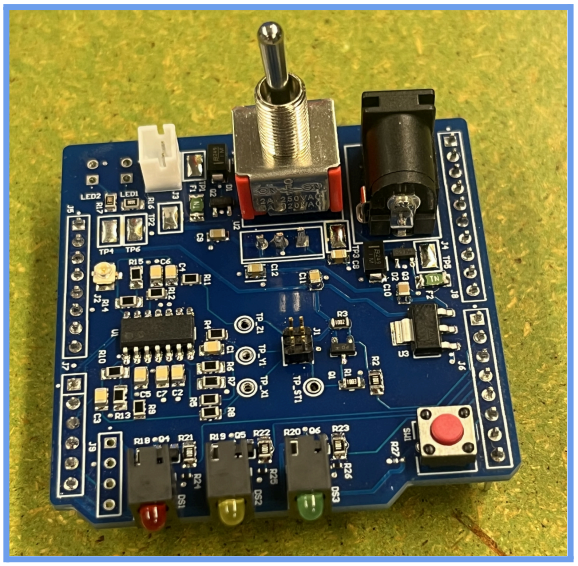


PCB Render - Side View

All component placement prioritises ease of use and understanding, as well as providing the user with a quality experience from the finished product. The placement of the button as well as the indicator LEDs and the tactile top mounted switch were all designed to reach the goal of providing a compact and efficient StepTracker.



PCB with SMD Components



PCB with all components

3. PCB Bill of Materials (BOM)

STEP TRACKER - PCB BILL OF MATERIALS

#	Ref Des	Qty	Description	Manufacturer	Manufacturer P/N	Supplier	Supplier P/N	Unit $ (AUD)	Total $ (AUD)
Capacitors									
1	C1–C7	7	MLCC 0.1µF 50V ±10% X7R 0805 SMD · Decoupling — power rails and Sallen-Key filter	Vishay	VJ0805Y104KXAAT	Mouser	77-VJ0805Y104KXAAT	$0.271	$1.90
2	C8, C11	2	MLCC 10µF 16V ±10% X5R 0805 SMD · Bulk decoupling — supply rails	Murata	GRM21BR61C106KE15L	Mouser	81-GRM21BR61C106KE15	$0.064	$0.13
3	C9	1	MLCC 1µF 25V ±10% X5R 0805 SMD · Regulator output filter	YAGEO	CC0805KKX5R8BB105	Mouser	603-CC805KKX5R8BB105	$0.104	$0.10
4	C10	1	MLCC 0.1µF 50V ±10% X7R 0805 SMD · Regulator input decoupling	YAGEO	CC0805KRX7R9BB104	Mouser	603-CC805KRX7R9BB104	$0.018	$0.02
5	C12	1	MLCC 1µF 10V ±10% X7R 0805 SMD · 3.3V regulator output filter	Samsung Electro-Mech.	CL21B105KPFNNNE	Mouser	187-CL21B105KPFNNNG	$0.044	$0.04
								Capacitors Subtotal	**$2.19**
Protection Devices									
6	D1, D2	2	TVS Diode 15V 24.4V Uni-Directional SMA · Input power rail transient protection	Bourns	SMAJ15A	Mouser	652-SMAJ15A	$0.173	$0.35
7	F1, F2	2	PPTC Resettable Fuse 0.25A 16V 1206 SMD · Overcurrent protection on power inputs	Eaton	PTS120616V025	Mouser	504-PTS120616V025	$0.161	$0.32
								Protection Subtotal	**$0.67**
Indicators / LEDs									
8	DS1	1	LED Green Diffused 3mm Right-Angle THT · Pace indicator — Running state	Kingbright	WP934CB/GD	Mouser	604-WP934CB/GD	$0.542	$0.54
9	DS2	1	LED Yellow Diffused 3mm Right-Angle THT · Pace indicator — Walking state	Kingbright	WP934CB/YD	Mouser	604-WP934CB/YD	$0.605	$0.61
10	DS3	1	LED Red Diffused 3mm Right-Angle THT · Pace indicator — Stationary state	Kingbright	WP934CB/ID	Mouser	604-WP934CB/ID	$0.579	$0.58
11	LED1, LED2	2	LED Super Red Diffused Round 3mm THT · Power supply rail indicators (+5V, +3V3)	Würth Elektronik	151031SS04000	Mouser	710-151031SS04000	$0.271	$0.54
								Indicators Subtotal	**$2.27**

#	Ref Des	Qty	Description	Manufacturer	Manufacturer P/N	Supplier	Supplier P/N	Unit $ (AUD)	Total $ (AUD)
Connectors									
12	J1	1	Conn Header Vert 6-pos 1.27mm · ADXL335 breakout board interface	Amphenol ICC	20021111-00006T4LF	Digi-Key	609-3710-ND	$0.740	$0.74
13	J2	1	RF Conn U.FL/IPEX Rcpt Straight 50Ω SMD · Z-axis filter BNC test output	Hirose	U.FL-R-SMT-1(10)	Mouser	798-U.FL-R-SMT-110	$1.910	$1.91
14	J3	1	Conn Header 2-pos 2mm JST PH Series · Battery connector (LiPo / 9V)	JST	B2B-PH-K-S(LF)(SN)	Mouser	306-B2BPHKSLFSNPP	$0.163	$0.16
15	J4	1	DC Power Jack 2.0mm ID 5.5mm OD 5A · Barrel jack — 9V external power input	Same Sky	PJ-102AH	Mouser	490-PJ-102AH	$0.962	$0.96
16	J5, J6	2	Conn Header Vert 8-pos 2.54mm · Arduino UNO R4 WiFi interface headers	Sullins	PREC008SAAN-RC	Digi-Key	S1012EC-08-ND	$0.467	$0.93
17	J7	1	Conn Header Vert 6-pos 2.54mm · Arduino UNO R4 WiFi interface header	Molex	22-28-4060	Mouser	538-22-28-4060	$0.444	$0.44
18	J8	1	Conn Header Vert 10-pos 2.54mm · Arduino UNO R4 WiFi interface header	Sullins	PREC010SAAN-RC	Digi-Key	S1012EC-10-ND	$0.590	$0.59
19	J9	1	Pass-Through Socket 4-pos 2.54mm · GC9A01 LCD SPI interface socket	Samtec	BCS-104-L-S-TE	Mouser	200-BCS104LSTE	$1.480	$1.48
								Connectors Subtotal	**$7.22**
Transistors / MOSFETs									
20	Q1	1	MOSFET P-Ch 50V 130mA SOT-23 · ADXL335 self-test ST pin driver	onsemi	BSS84LT1G	Digi-Key	BSS84LT1GOSCT-ND	$0.276	$0.28
21	Q2, Q3	2	MOSFET P-Ch 30V 3.8A SOT-23 · High-side power switching	Diodes Inc.	DMP3099L-7	Digi-Key	DMP3099L-7DICT-ND	$0.303	$0.61
22	Q4, Q5, Q6	3	MOSFET N-Ch 60V 300mA SOT-23 · Low-side switching	Infineon	2N7002H6327XTSA2	Digi-Key	2N7002H6327XTSA2CT-ND	$0.214	$0.64
								Transistors Subtotal	**$1.52**

#	Ref Des	Qty	Description	Manufacturer	Manufacturer P/N	Supplier	Supplier P/N	Unit $ (AUD)	Total $ (AUD)
Resistors									
23	R1	1	Res SMD 100kΩ ±1% 1/8W 0805 · ADXL ST pin pull-down to GND	YAGEO	AC0805FR-07100KL	Digi-Key	YAG3688CT-ND	$0.055	$0.06
24	R2	1	Res SMD 1kΩ ±1% 1/8W 0805 · MOSFET gate series protection (Arduino D6)	YAGEO	AC0805FR-071KL	Digi-Key	YAG3703CT-ND	$0.055	$0.06
25	R3	1	Res SMD 10kΩ ±1% 1/8W 0805 · MOSFET gate pull-up to +3V3	YAGEO	RC0805FR-1010KL	Digi-Key	13-RC0805FR-1010KLCT-ND	$0.047	$0.05
26	R4, R5, R8, R9, R11, R12	6	Res SMD 16kΩ ±1% 1/4W 0805 · Sallen-Key LPF series resistors (all 3 axes)	Vishay	CRCW080516K0FKEA	Digi-Key	541-16.0KCCT-ND	$0.040	$0.24
27	R6, R10, R14	3	Res SMD 5.62kΩ ±1% 1/4W 0805 · Sallen-Key gain resistor Rf (all 3 axes)	Vishay	CRCW08055K62FKEA	Digi-Key	541-5.62KCCT-ND	$0.049	$0.15
28	R7, R13, R15	3	Res SMD 10kΩ ±1% 1/4W 0805 · Sallen-Key gain resistor Rg (all 3 axes)	Vishay	CRCW080510K0FKEA	Digi-Key	541-10.0KCCT-ND	$0.049	$0.15
29	R16	1	Res SMD 300Ω ±1% 1/8W 0805 · LED current-limiting resistor	Bourns	CR0805-FX-3000ELF	Digi-Key	CR0805-FX-3000ELFCT-ND	$0.055	$0.06
30	R17	1	Res SMD 130Ω ±1% 1/8W 0805 · LED current-limiting resistor	Bourns	CR0805-FX-1300ELF	Digi-Key	118-CR0805-FX-1300ELFCT-ND	$0.055	$0.06
31	R18, R19, R20	3	Res SMD 220Ω ±1% 1/4W 0805 · LED current-limiting resistors	Vishay	CRCW0805220RFKEA	Digi-Key	541-220CCT-ND	$0.049	$0.15
32	R21, R22, R23	3	Res SMD 1kΩ ±1% 1/8W 0805 · General purpose / pull resistors	YAGEO	RC0805FR-071KL	Digi-Key	311-1.00KCRCT-ND	$0.035	$0.11
33	R24–R27	4	Res SMD 10kΩ ±1% 1/16W 0402 · General purpose pull resistors	YAGEO	RC0402FR-0710KL	Digi-Key	311-10.0KLRCT-ND	$0.022	$0.09
								Resistors Subtotal	**$1.14**
Switches									
34	S1	1	Toggle Switch DPDT 5A 120VAC Panel Mount · Main power on/off switch	-	ANT23SECQE	Digi-Key	2449-ANT23SECQE-ND	$5.460	$5.46
35	SW1	1	Tactile Switch SPST-NO 0.05A 12V THT · User interface button (Arduino D2)	-	FSM2JART	Digi-Key	450-FSM2JARTCT-ND	$0.270	$0.27
								Switches Subtotal	**$5.73**
Test Points									
36	TP1–TP6	6	SMD Test Point 3.4×1.8mm · Voltage and signal probe access points	-	-	-	-	-	-
37	TP_ST1, TP_X1, TP_Y1, TP_Z1	4	Test Point Loop 1mm	-	-	-	-	-	-

#	Ref Des	Qty	Description	Manufacturer	Manufacturer P/N	Supplier	Supplier P/N	Unit $ (AUD)	Total $ (AUD)
Integrated Circuits									
38	U1	1	IC Op-Amp General Purpose Quad 14-SOIC · Sallen-Key LPF active element (3 channels)	Microchip Technology	MCP6004-I/SL	Digi-Key	MCP6004-I/SL-ND	$0.800	$0.80
39	U2	1	IC Voltage Reg Linear 5V 1.5A TO-220-3 · +5V supply regulator from Vin	Texas Instruments	LM7805CT/NOPB	Digi-Key	296-47192-ND	$2.660	$2.66
40	U3	1	IC Voltage Reg Linear 3.3V 800mA SOT-223 · +3V3 supply for ADXL335	STMicroelectronics	LD1117S33TR	Digi-Key	497-1242-1-ND	$0.363	$0.36
								ICs Subtotal	**$3.82**
PCB Manufacturing									
41	PCB	1	Custom 2-Layer FR4 PCB · 1.6mm, HASL, 2oz Cu, green silkscreen, Step Tracker Shield	JLCPCB	Custom order	JLCPCB	jlcpcb.com	$18.000	$18.00
42	Stencil	1	SMD Solder Paste Stencil — Top Side · Stainless steel, framed, for SMD assembly	JLCPCB	Custom order	JLCPCB	jlcpcb.com	$12.000	$12.00
								PCB Manufacturing Subtotal	**$30.00**
Accessories & Peripheral Components									
43	—	1	Battery Connector Snap 9V with Leads 150mm · Connects 9V battery to J3 JST connector	Keystone	00091	Digi-Key	1568-00091-ND	$2.660	$2.66
44	—	1	BNC Male to U.FL Female Adapter Cable 75mm · Connects J2 to oscilloscope BNC input	Bevotop	BT1488	AliExpress	BT1488	$4.490	$4.49
45	—	1	JST PH Housing 2-pos · Mating connector for J3 battery header	JST	PHR-2	Mouser	306-PHR-2PP	$0.077	$0.08
46	—	2	JST PH Crimp Terminal ×2 · Mating crimps for PHR-2 battery housing	JST	SPH-002T-P0.5S	Mouser	306-SPH-002T-P0.5S	$0.062	$0.12
47	—	1	9V Alkaline Battery 4-Pack · Primary power source — bench testing / initial deployment	Energizer	UPN-169922	Woolworths	710051	$6.250	$6.25
								Accessories Subtotal	**$13.60**
TOTAL PCB BOM COST (AUD) · Per-system pricing · Includes PCB manufacturing and accessories									**$68.17**

PCB Testing and Validation

1. PCB subsystem testing & validation

1.1 Test overview

All PCB subsystem testing and validation procedures were undertaken at ambient temperature (24.5℃) with the Arduino Uno R4 receiving constant power through the USB-C connection. The 5V rail measured 4.67V using the USB supply. The 9V battery was not used for the majority of the tests as the USB-C connection provides more consistent results. The 9V battery was only used for testing the JST and Barrel jack power connectors.

Each subsystem was tested independently against our schematic design calculations and part manufacturer datasheets. Tests are grouped by subsystem and results are recorded in the testing tables below.

1.2 Test Equipment

The tests undertaken in the validation procedure requires various electronic measurement apparatus, including;

- Oscilloscope and leads
- uFL to BNC connector
- Multimeter and leads
- USB-C power supply
- DC Barrel Jack Power supply

1.3 Visual Inspection

Test ID	Item	Acceptance Criteria	Result	Notes
VI-01	Solder joint quality	No missing or shorted joints between pads visible under microscope	Pass - See media in appendix	-
VI-02	Component Orientation	All ICs, LEDs and MOSFETs and Diodes are correctly oriented with silkscreen markings	Pass - see media in appendix	-
VI-03	Component Population	All line items in the BOM have been placed on the board. No empty pads with component designators.	Pass - see media in appendix	-
VI-04	PCB Cleanliness	No flux residue, splattered solder or other debris present on the board.	Pass - see media in appendix	-
VI-05	PCB Mechanical Condition	No delamination between board layers, no lifted pads or damaged traces.	Pass - see media in appendix	-

1.4 Power Supply Subsystem

Test ID	Parameter	Test Condition	Specification	Measured Value	Result	Notes
P-01	Test toggle switch	Confirm switch turns on and off pcb	Indicator lights should turn on and off	N/A	Pass	-
P-02	Pre-power short circuit check. 5V to GND	No Power input, multimeter in continuity mode and test between 5V and GND	Open Loop reading on multimeter	Open Loop	Pass	Perform this test before any power input.
P-03	Pre-power short circuit check. 3V3 to GND	No Power input, multimeter in continuity mode and test between 3V3 and GND	Open Loop reading on multimeter	Open Loop	Pass	Perform this test before any power input.
P-04	5V rail voltage with no load	USB-C connected with no code running	4.50–5V	4.74V	Pass	-
P-05	5V rail voltage with full load	USB-C connected with code running	4.50-5V	4.67V	Pass	-
P-06	3V3 rail voltage at ADXL335 VS pin	System running with full functionality	3.00 to 3.60 V	3.310	Pass	-
P-07	5V indicator LED current draw	Toggle switch ON, + 5V LED illuminated	<15mA	10mA	Pass	LED illuminates correctly
P-08	3V3 indicator LED current draw	Toggle switch ON, +3V3 LED illuminated	<15mA	10mA	Pass	LED illuminates correctly
P-09	Supply voltage at 5V components	System running, measure at MCP6004 VS, GC9A01 VCC, Arduino VCC	4.50–5V	4.50-5V	Pass	5V components receiving adequate voltage.
P-10	Total system current draw	System powered at 9V through JST connector with multimeter in series	<500mA	210mA	Pass	Baseline current draw to estimate battery life.

1.5 Accelerometer Subsystem

Test ID	Parameter	Test Condition	Specification	Measured Value	Result	Notes
A-01	XOUT zero-g bias voltage	Board flat on level surface (0 g on X axis). Measure at ADXL335 XOUT pin	1.6V +/- 0.15V	1.62V	Pass	Values outside this range may indicate ADXL335 damage.
A-02	YOUT zero-g bias voltage	Board flat on level surface (0 g on Y axis). Measure at ADXL335 YOUT pin	1.6V +/- 0.15V	1.64V	Pass	Values outside this range may indicate ADXL335 damage.
A-03	ZOUT zero-g bias voltage	Board standing on X edge (0 g on Z axis). Measure at ADXL335 ZOUT pin	1.6 +/- 0.3V	1.66V	Pass	Values outside this range may indicate ADXL335 damage.
A-04	X-axis sensitivity (1g - 0g bias)	Board tilted 90° to place X axis vertical (1 g acting on X). Measure XOUT.	297–363 mV/g	324 mV/g	Pass	ADXL335 sensitivity is ratiometric. At VS = 3 V the datasheet specifies 270–330 mV/g. At VS = 3.3 V this scales to approximately 297–363 mV/g.
A-05	Y-axis sensitivity (1g - 0g bias)	Board tilted 90° to place Y axis vertical (1 g acting on Y). Measure YOUT.	297–363 mV/g	322 mV/g	Pass	See A-04
A-06	Z-axis sensitivity (1g - 0g bias)	Board flat on level surface (1 g acting on Z). Measure ZOUT.	297–363 mV/g	322 mV/g	Pass	See A-04
A-07	Self Test MOSFET (BSS84) gate voltage - off state	Normal firmware operation. D6 HIGH. Measure voltage at MOSFET gate	4.5-5V	4.55V	Pass	Confirms MOSFET is being pulled to 5V by Arduino digital pin.
A-08	Self Test	Self-test active.	0.00-0.30V	0.077V	Pass	Confirms activating

Test ID	Parameter	Test Condition	Specification	Measured Value	Result	Notes
	MOSFET (BSS84) gate voltage - on state	D6 LOW. Measure voltage at MOSFET gate.				the Self Test pin pulls the gate voltage below 0.3V.
A-09	Self Test Response - X Axis	Trigger Self test through LCD menu. Read X-axis mV delta from LCD display.	−833 to −357 mV	−512 mV	Pass	Confirms that the ADXL335 signal is being read correctly from the Arduino ADC.
A-10	Self Test Response - Y Axis	Same self-test event as A-09. Read Y-axis mV delta from LCD display	+357 to +833 mV	+566 mV	Pass	See A-09
A-11	Self Test Response - Z Axis	Same self-test event as A-09. Read Z-axis mV delta from LCD display	+600 to +1416 mV	+1112 mV	Pass	See A-09

1.6 Analog Filter Subsystem

Filter Design Parameters

Parameter	Theoretical Value	Simulated Value
Filter Cutoff Frequency	99.5 Hz	100 Hz
Filter gain	1.586 V	1.590 V

Test ID	Parameter	Test Condition	Amplitude	Phase	Result	Notes
F-01	Passband Signal	10 Hz, 1 Vpp, 1.5 V DC offset	1.59Vpp	-8 degrees	Pass	Passband Gain and Phase shift are within 0.5% of theoretical values.
F-02	Cutoff frequency signal	100 Hz, 1 Vpp, 1.5 V DC offset	1.15Vpp	-90 degrees	Pass	Measured -3dB = 1.15Vpp. Theoretical -3dB = 1.124dB, within 2.5%.
F-03	Stopband Signal	1000 Hz, 1 Vpp, 1.5 V DC offset	0.035 Vpp	-171 degrees	Pass	-

1.7 Faults and Corrective Actions

One fault was located on the PCB. The third leg of the DC Barrel Jack was incorrectly connected and was required to be grounded. The result of this mistake was zero power coming through the Barrel Jack. The fix for this was a small jumper lead between the unconnected leg of the Barrel Jack to the GND leg. This resolved the problem and restored the PCB to its correct and operational state.

1.8 Test Summary

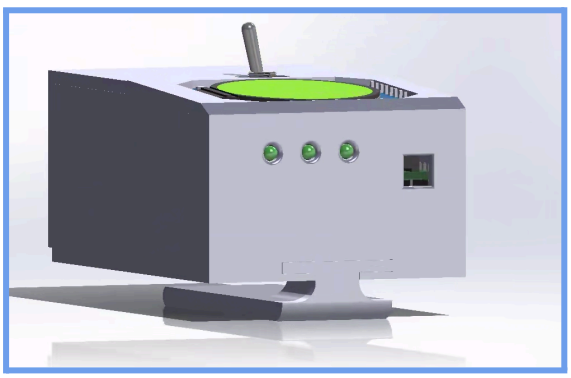
In conclusion, the testing and validation process undertaken on the StepTracker successfully verified its operational integrity, as well as proving the systems signal accuracy against the designed specifications. Through a rigorous testing procedure, the PCB was audited first with a multi stage visual inspection and pre power continuity checks, then an active load test of both the 5V and 3.3V power rails.

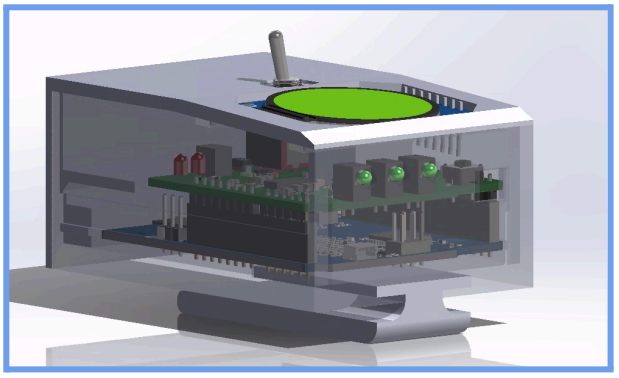
Testing with the accelerometer validated that it was operating within its expected parameters from the relevant datasheet, and proving it responded correctly to the hardware activated self test procedure.

The further frequency response testing of the filtering circuitry verified that the finished product met the design specifications, achieving the targeted passband gain and a clean 100Hz cutoff frequency.

Ultimately, the results confirm that the hardware platform is stable and safe, and optimised for a quality user experience.

Final Enclosure Design (MCAD)





Overview

The StepTracker system enclosure consists of four 3d printed components. The StepCase houses the 2 board PCB stack on a running rail for easy access and disassembly. It also has dedicated front cutouts for the indicator LEDs and user button.

The LCD screen is held in by the StepRoof which has a window for angled LCD viewing. The RearCase is a removable back panel that houses the battery for the system as well as a dedicated cutout for the toggle switch.

Finally, the BeltCase is a friction fit removable clip that allows the user to utilise the system in 2 different modes.

The entire enclosure design is built to be assembled directly off the 3D print bed and requires no tools or fasteners to put together. All parts have been tolerance checked and fit together with snug friction fitment. It is entirely printed from PLA + on a Bambu Labs A1 mini.

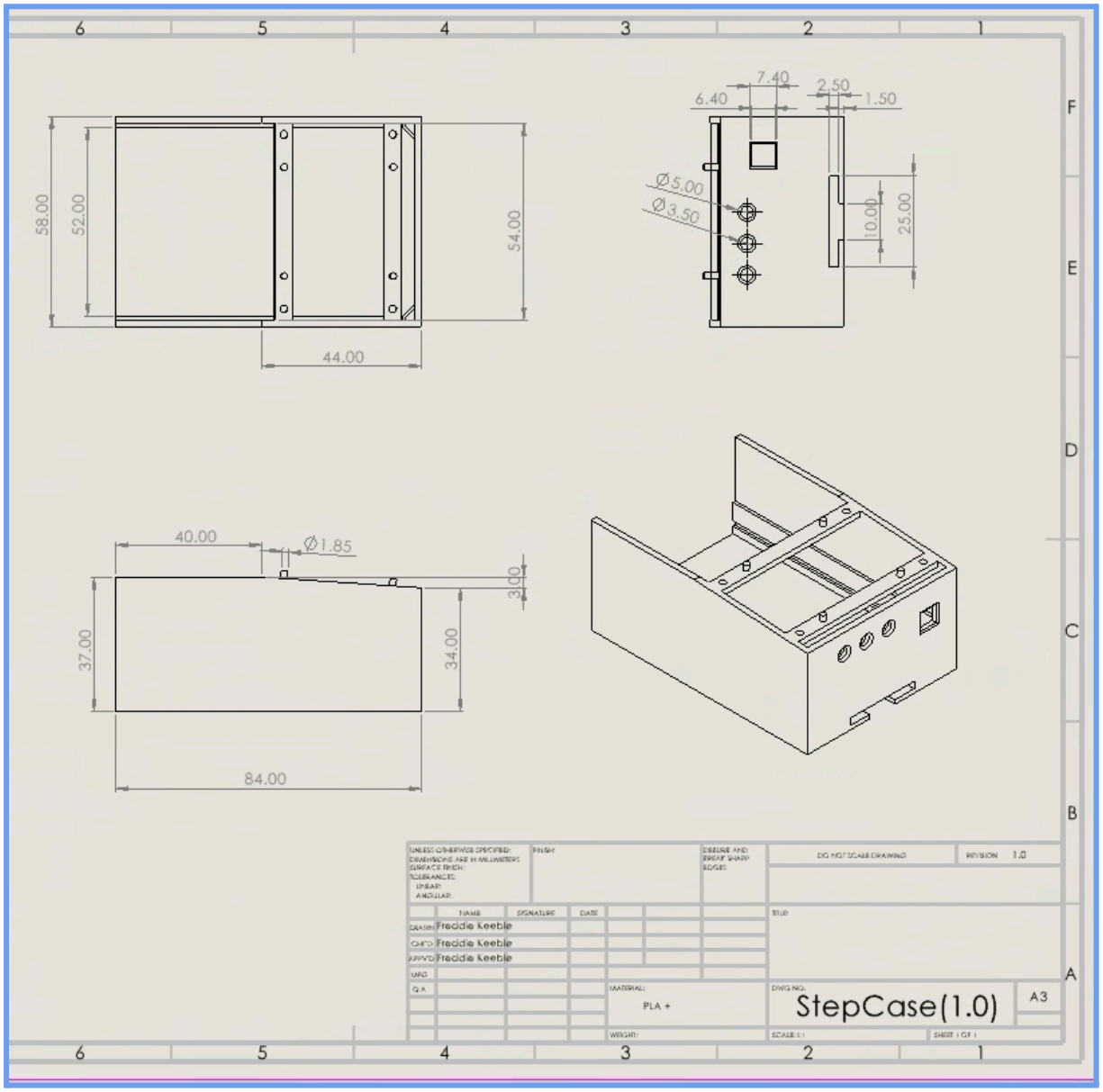
Assembly instructions

To assemble the steptracker, slide the assembled PCB stack into the Step Case from the rear opening along the rail. Then align the mount holes on the LCD screen with the pins on top of the StepCase. Once the screen is in place, align the Step Roof with the outside of the screen and apply light pressure to push into place. Finally, insert the 9V battery into its slot in the RearCase and then slide the RearCase along the keyed groove until it clicks closed.

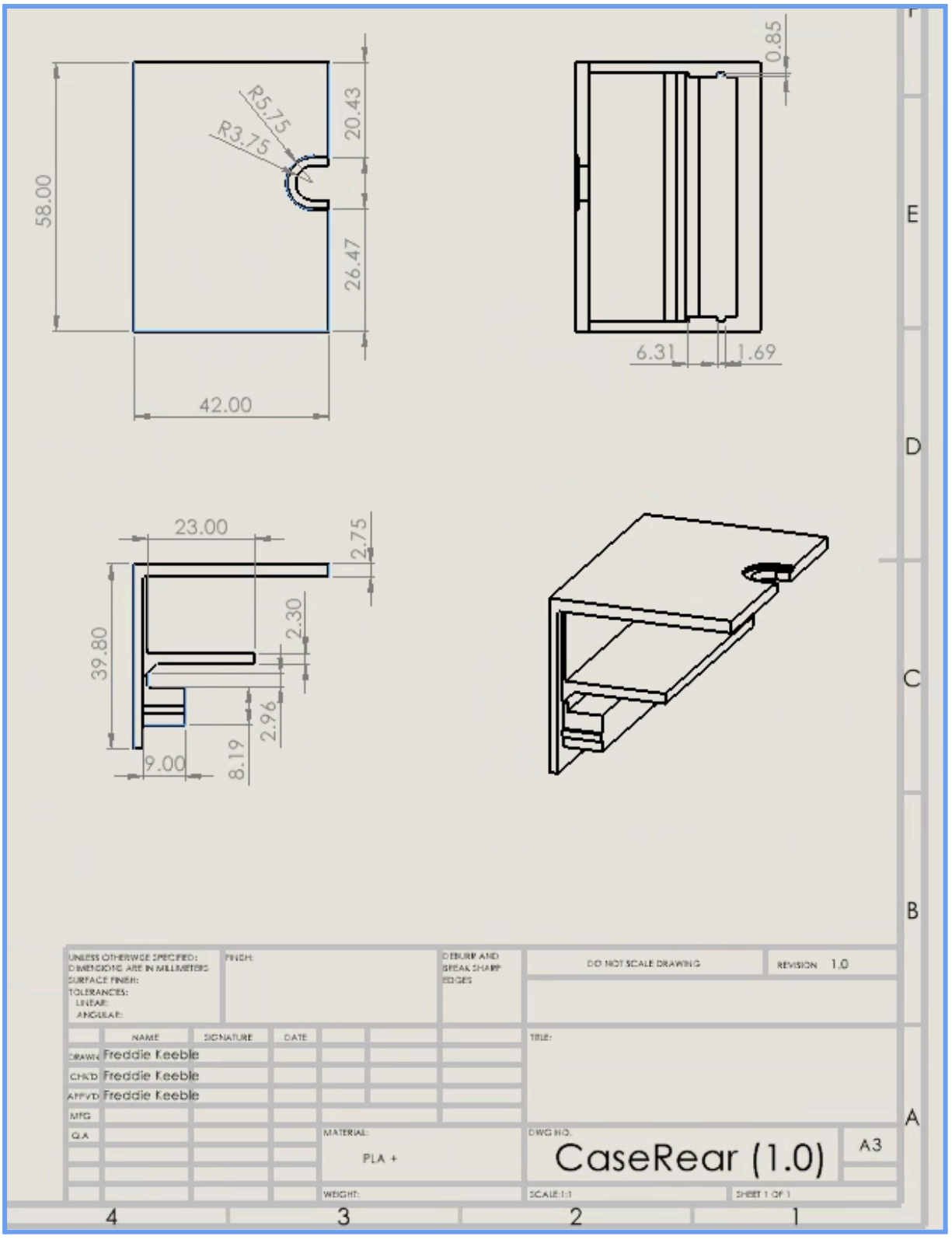
1. Technical drawings of all parts

Mechanical drawings have been cropped to allow for viewing in portrait PDF format.

Step Case (1.0)



CaseRear (1.0)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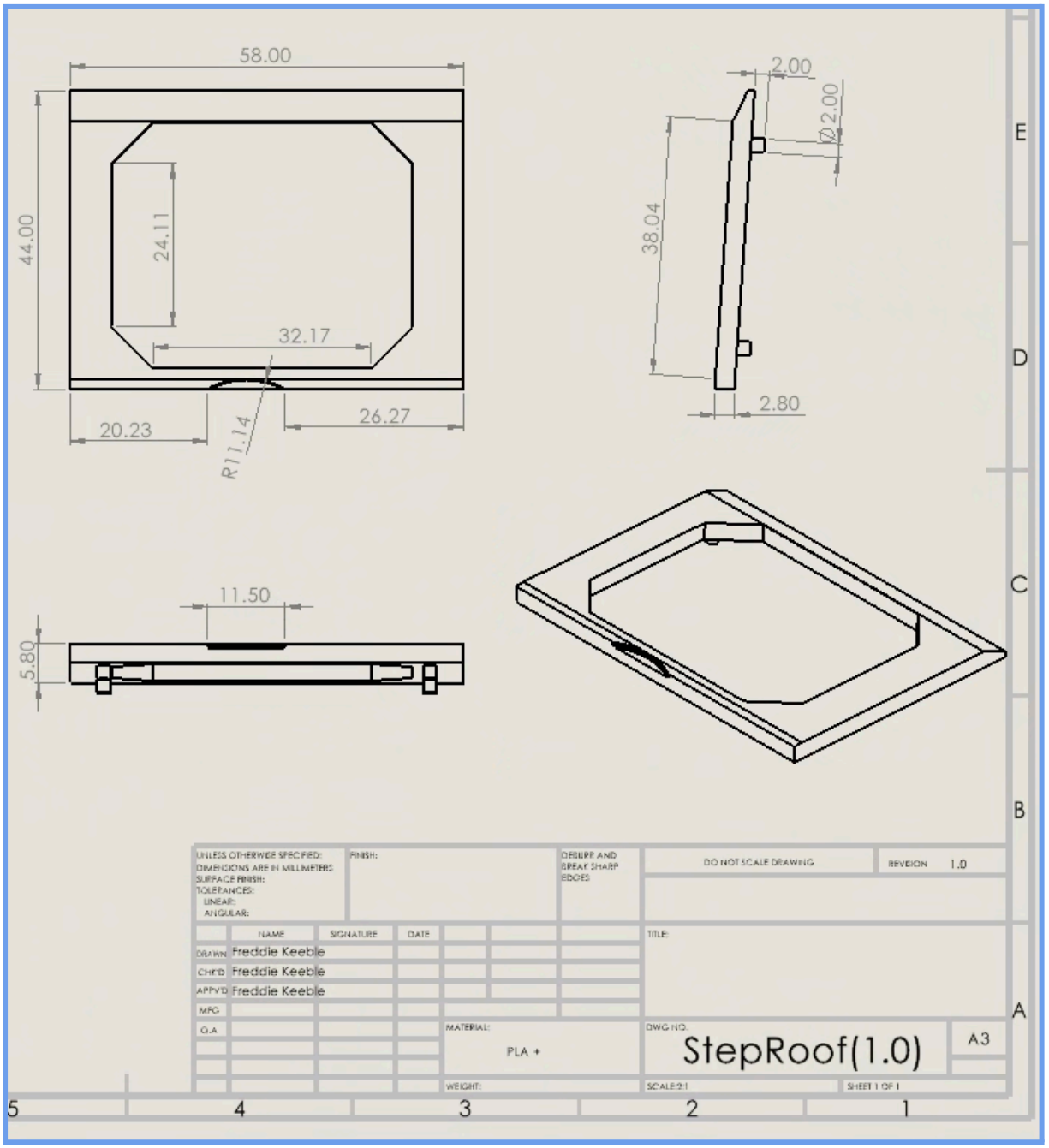
StepRoof (1.0)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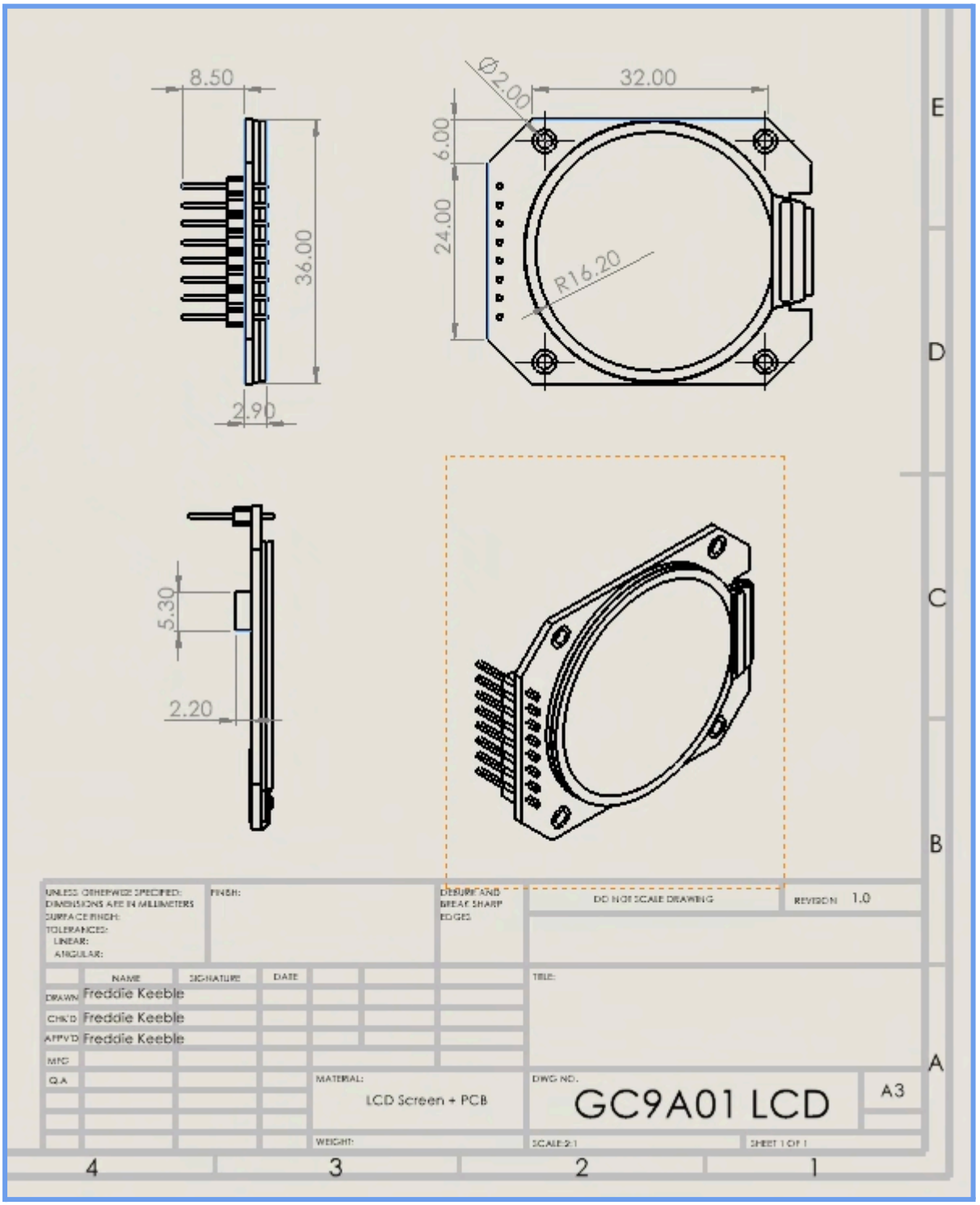
GC9A01 LCD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Ø2.00
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E
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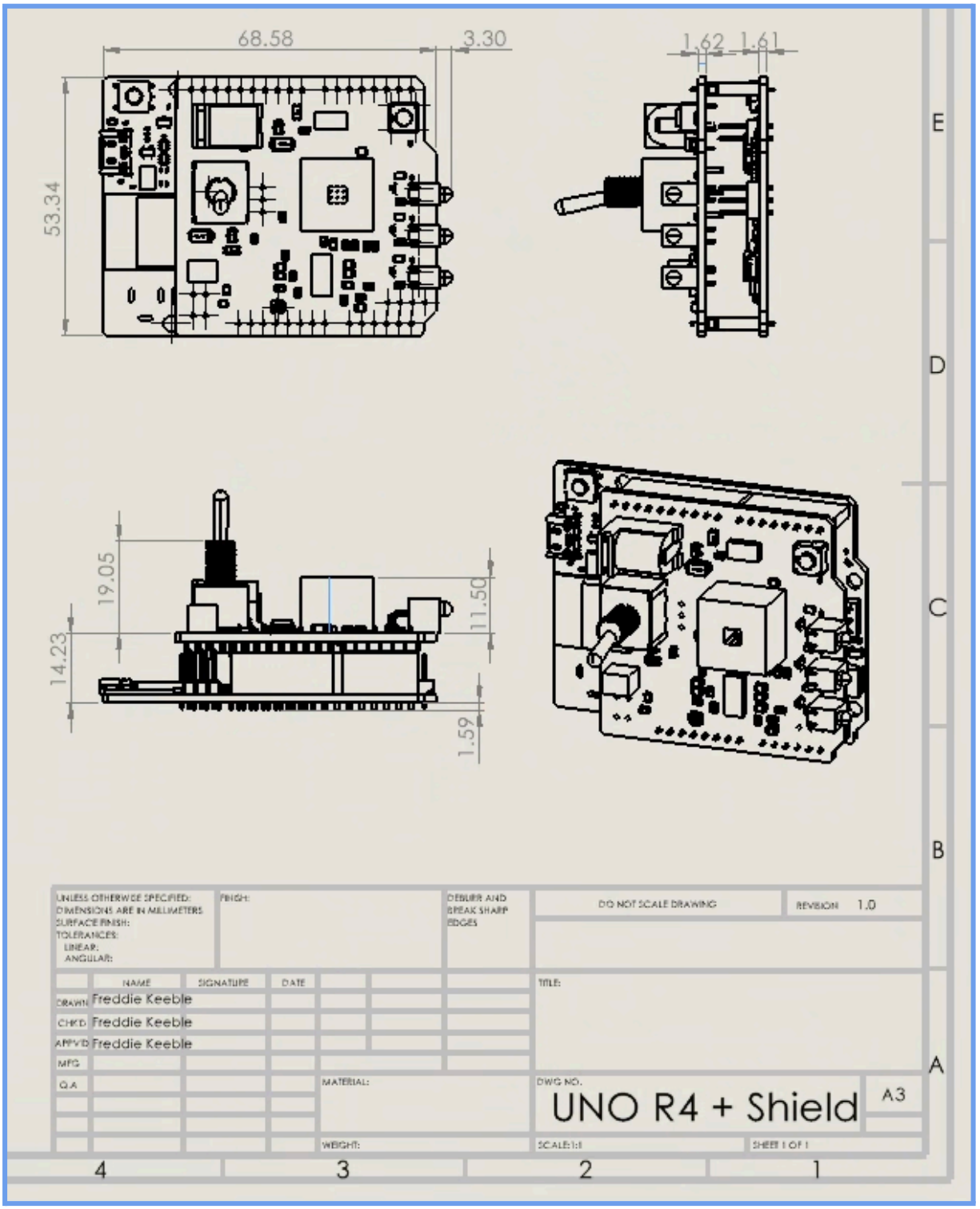
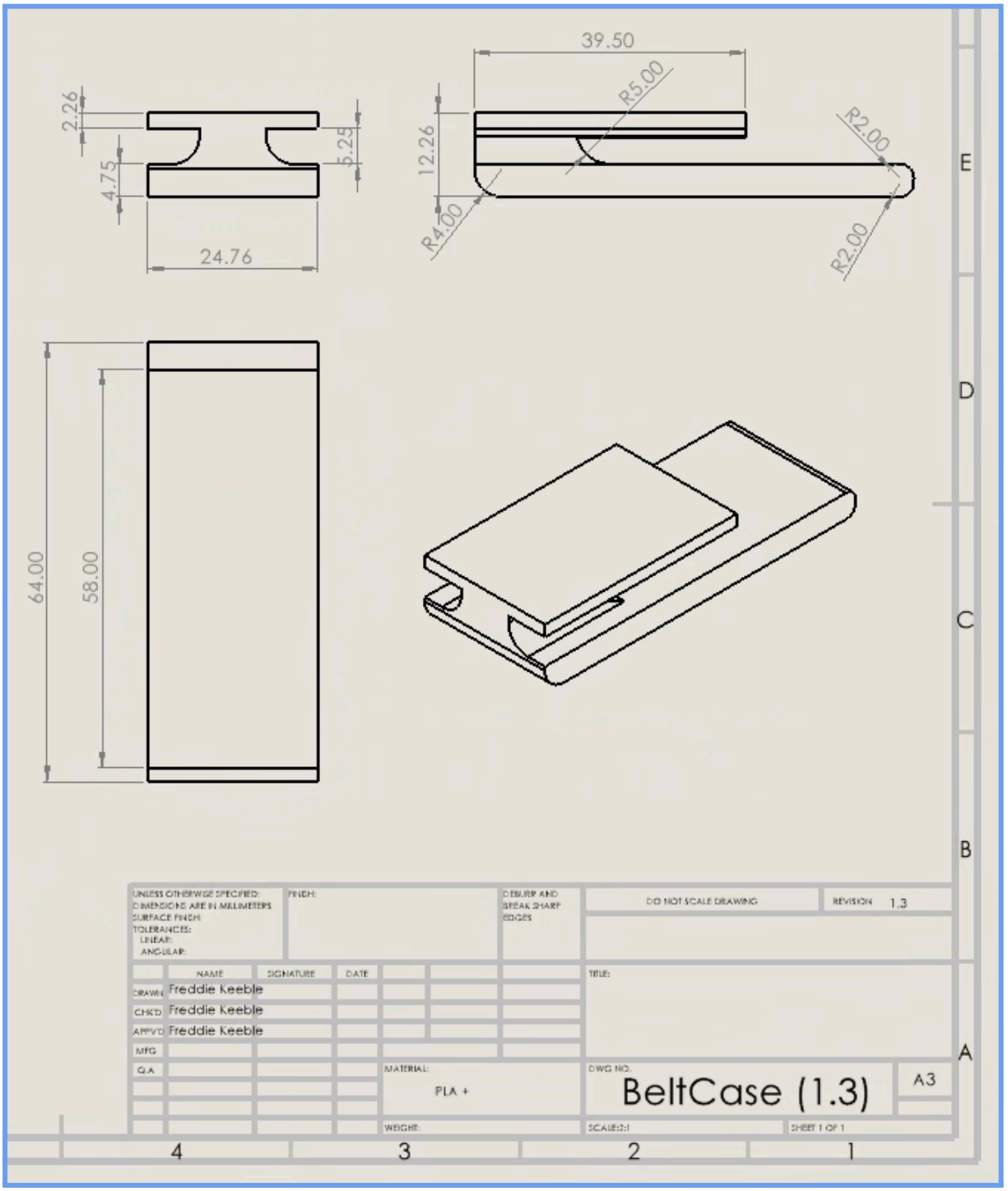
UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:
FINISH:
DEBURR AND
BREAK SHARP
EDGES
DO NOT SCALE DRAWING
REVISION 1.0
NAME
SIGNATURE
DATE
TITLE:
DRAWN Freddie Keeble
CHK'D Freddie Keeble
APPV'D Freddie Keeble
MFG
Q.A
MATERIAL:
LCD Screen + PCB
DWG NO.
GC9A01 LCD
A3
WEIGHT:
SCALE:2:1
SHEET 1 OF 1
4
3
2
1

UNO R4 and Custom Step Tracker Shield



Belt Case (1.3)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R5.00
R2.00
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R4.00
R2.00
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UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:
FINISH:
DEBURR AND BREAK SHARP EDGES
DO NOT SCALE DRAWING
REVISION 1.3
NAME
SIGNATURE
DATE
DRAWN Freddie Keeble
CHK'D Freddie Keeble
APPV'D Freddie Keeble
MFG
Q.A
TITLE:
MATERIAL:
PLA +
DWG NO.
BeltCase (1.3)
A3
WEIGHT:
SCALE:2:1
SHEET 1 OF 1
E
D
C
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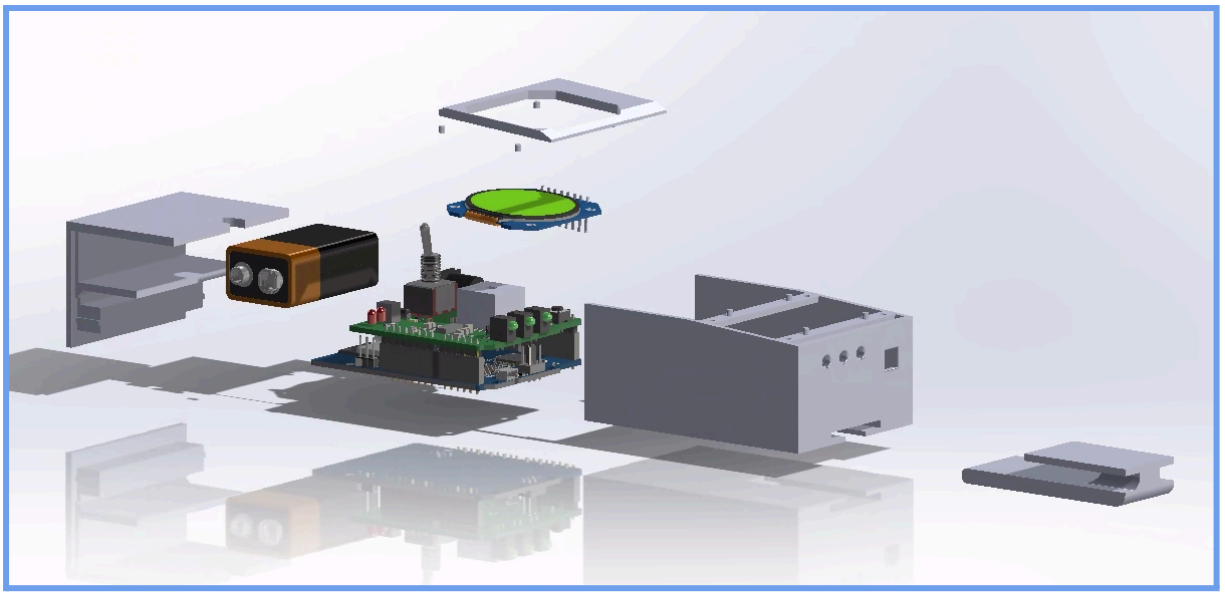
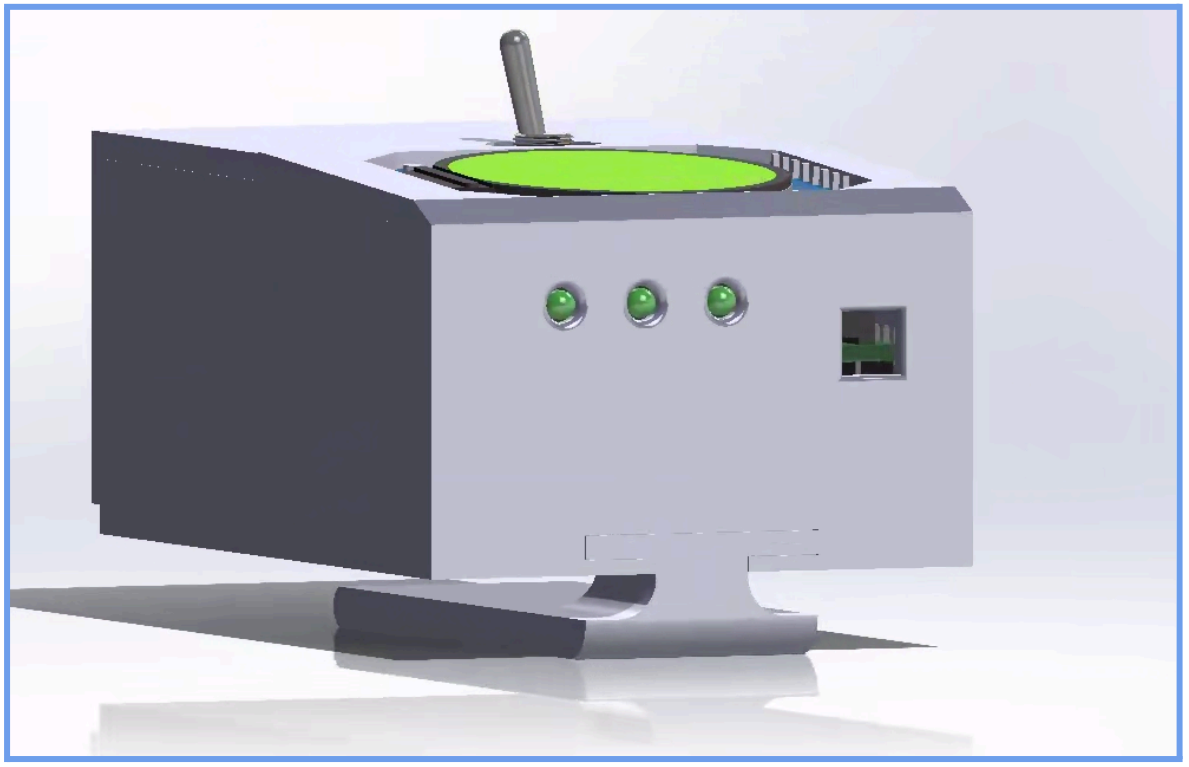
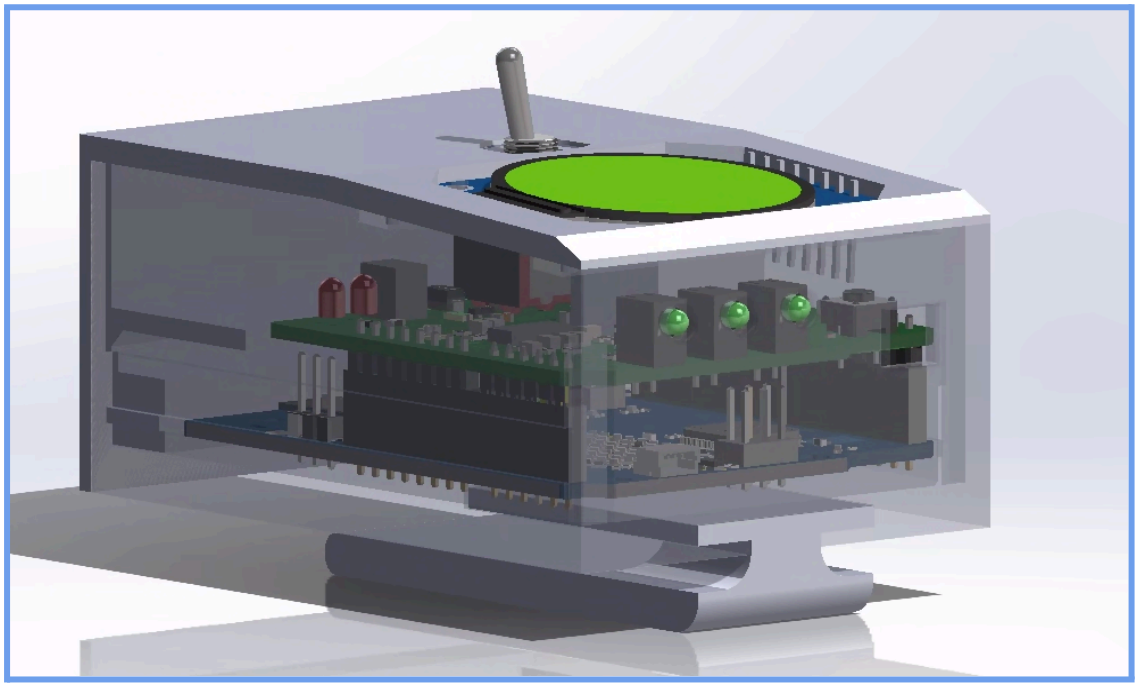
1

2. Exploded view including ECAD parts



3. 3D models and 3D assembly including ECAD parts





4. Bill of Materials (BOM)

STEP TRACKER — COMPLETE SYSTEM BILL OF MATERIALS

#	Item / Assembly	Qty	Supplier	Part / SKU	Unit $ (AUD)	Total $ (AUD)
Electronic Assemblies & Modules						
1	Step Tracker Shield — Assembled PCB	1	In-house assembly	See PCB BOM	$68.18	$68.18
2	Arduino UNO R4 WiFi	1	Core Electronics	CE09127	$39.95	$39.95
3	GC9A01 1.28" Round TFT LCD — Breakout Board	1	AliExpress / Waveshare	GC9A01-Module	$12.50	$12.50
4	ADXL335 Accelerometer — Breakout Board	1	UTS	DEV-09269	$8.50	$8.50
					Electronics Subtotal	**$129.13**
Power Supply						
5	9V Lithium Battery	1	Jaycar / Woolworths	LS22	$19.95	$19.95
					Power Subtotal	**$19.95**
Enclosure (3D Printed — PLA)						
6	Enclosure Base	1	In-house 3D print	MCAD-BASE-01	$2.50	$2.50
7	Enclosure Top	1	In-house 3D print	MCAD-TOP-01	$1.50	$1.50
8	Enclosure Rear + Belt Clip Mount	1	In-house 3D print	MCAD-REAR-01	$0.75	$0.75
					Enclosure Subtotal	**$4.75**
TOTAL SYSTEM COST (AUD) · Complete Step Tracker system · Per-unit pricing						**$153.83**

Final Embedded Software

1. User Guide and Documentation of the Step Tracker Program

1.1 Software Architecture Overview

The Step Tracker firmware is written in C/C++ with the Arduino framework and runs on the Arduino UNO R4 WiFi. The code is split into eight files so each part is easy to find and edit. All pin numbers, settings, colours and shared variables live in one header file, config.h, so the whole system can be adjusted from one place.

The program uses a simple non-blocking main loop. After a one-time setup and calibration, loop() repeats about every 20 ms. Reading the sensor, detecting steps and working out pace happen every loop, no matter what is on screen, so the step count keeps going even in the menu or on a clock screen.

External libraries used:

Library	Purpose
Arduino_GFX_Library	Drawing functions for the GC9A01 240×240 round LCD.
WiFiS3	Controls the UNO R4 WiFi's on-board Wi-Fi.
WiFiUdp	UDP used by the time client.
NTPClient	Gets network time for the clock screens.
math.h	Square-root and trig functions for the magnitude and tilt maths.

Source file responsibilities:

File	Responsibility
config.h	Pin map, settings, colours, states and all shared variables.
StepTracker.ino	Main file: globals, setup(), loop(), button handling and menu/screen navigation.
Calibration.ino	Guided three-position calibration and its on-screen steps.
SelfTest.ino	ADXL335 self-test and pass/fail result.
StepCounter.ino	Step detection from the acceleration magnitude.
PaceIdentification.ino	Works out pace: stationary, walking or running.
LCD.ino	All screen drawings: steps, clocks, attitude indicator, Wi-Fi icon and menu.
Wireless.ino	Wi-Fi connection handling and network time updates.

1.2 Program Structure

- **setup()** runs once at power on. It starts the serial port at 115200 baud for debug messages, sets up the button, LED, self-test and display pins, turns the LEDs off and holds the self-test pin HIGH (self-test off). It then starts the GC9A01 display, clears the screen and runs calibration. After calibration it opens the menu instead of going straight to the step screen.

- **loop()** runs over and over. Each pass it:
    1. Checks the button with checkButton()
    2. Runs updateStepDetection() and updatePaceState() so motion is always tracked
    3. Updates Wi-Fi when it is on
    4. Applies the step-count display logic
    5. Redraws the active screen if the menu is closed

  A delay(20) at the end sets the loop speed.

- **Step-count display.** The detector adds 1 to an internal steps counter for each impact it sees. A fast runner often makes only one clear impact per pair of steps, so the loop keeps a separate displayStepCount that adds 2 when steps are close together (interval < 700 ms i.e. running) and 1 otherwise. This is the adaptive step counting noted on the front page.

1.3 Configuration and Global State (config.h)

Pin assignments

Signal	Arduino Pin	Macro	Direction
Mode / menu button	D2	BUTTON_PIN	Input
Accelerometer X output	A2	ACCEL_X_PIN	Analog in
Accelerometer Y output	A1	ACCEL_Y_PIN	Analog in
Accelerometer Z output	A0	ACCEL_Z_PIN	Analog in
Red LED (running / stationary)	D5	LED_RED_PIN	Output
Yellow LED (walking)	D4	LED_YELLOW_PIN	Output
Green LED (running)	D3	LED_GREEN_PIN	Output
LCD data/command	D9	TFT_DC	Output
LCD chip select	D8	TFT_CS	Output
LCD reset	D10	TFT_RST	Output
ADXL335 self-test (via MOSFET)	D6	ST_PIN	Output

Key tuning constants

Constant	Value	Meaning
SAMPLES	100	ADC samples averaged per orientation during calibration.
STEP_THRESHOLD_HIGH	1.4 g	Upper hysteresis threshold that arms a step.
STEP_THRESHOLD_LOW	0.70 g	Lower hysteresis threshold that confirms a step.
STEP_MIN_INTERVAL_MS	250 ms	Shortest accepted interval between steps (debounces false peaks).
STEP_MAX_INTERVAL_MS	3000 ms	Idle timeout after which motion is treated as stationary.

Constant	Value	Meaning
CADENCE_RUNNING_MIN	140 spm	Cadence at or above which pace is classified as running.
LONG_PRESS_MS	2000 ms	Hold time that distinguishes a long press from a short press.
DEBOUNCE_MS	50 ms	Minimum hold time for a valid short press.
NTP_UTC_OFFSET_SEC	36000 s	UTC+10 offset (AEST) applied to network time.

1.4 Subroutine Reference

Every subroutine is listed below, grouped by file. Related draw/update pairs share a row.

Main control - StepTracker.ino

Function	Description
setup()	Initialises serial, GPIO and the display, runs calibration, then opens the menu.
loop()	Cooperative scheduler (~20 ms): button, step detection, pace, Wi-Fi and screen refresh.
checkButton()	Times the button - long press (≥ 2 s) opens the menu / selects; short press cycles screens or menu items.
cycleScreen()	Advances to the next screen (clock screens included only when Wi-Fi is on).
selectMenuItem()	Runs the chosen menu action (self-test, calibration or Wi-Fi) and returns to the step screen.
drawCurrentScreen()	Full redraws, whichever screen is currently active.

Calibration - Calibration.ino

Function	Description
runCalibration()	Guided three-orientation routine; computes per-axis offsets and sensitivities and displays them.
takeSamples(orientation)	Averages 100 ADC reads per axis with an on-screen progress animation.
showStepInstruction(step)	Shows the prompt, target axis values, graphic and LED for one orientation.
showSampling(step)	Shows the “Sampling…” screen during acquisition.
showResults()	Displays the computed offsets and sensitivities for 3 s.
drawBoardGraphic(step)	Draws the board-orientation icon with an axis arrow.
drawProgressDots(currentStep)	Three progress dots: done (green), current (white), pending (grey).
waitForButton()	Blocks until a debounced press-and-release.
setLEDs(r, y, g)	Drives the red, yellow and green indicator LEDs.
printCentred(text, y, colour, size)	Prints horizontally-centred text at a given position.

Self-test - SelfTest.ino

Function	Description
runSelfTest()	Asserts the ADXL335 ST line, converts the ADC deltas to g, checks per-axis pass bands (X: -1.51 to -0.65 g, Y: +0.65 to +1.51 g, Z: +1.10 to +2.56 g) and shows PASS/FAIL.

Step counting - StepCounter.ino

Function	Description
updateStepDetection()	Magnitude hysteresis (arm > 1.4 g, confirm < 0.70 g) with 250-3000 ms interval gating; counts steps and stores the interval. Skipped during self-test.

Pace identification - PaceIdentification.ino

Function	Description
updatePaceState()	Sets stationary / walking / running from cadence (60000 / interval; ≥ 140 spm = running). Uses the stored interval to stay flicker-free.

Display - LCD.ino

Function	Description
drawStepScreen() / updateStepScreen()	Draw and incrementally refresh the step count, pace label, Wi-Fi glyph and pace LEDs.
drawAnalogClock() / updateAnalogClock()	Analog clock face and hands driven from NTP time; handle the no-Wi-Fi case.
drawClockFace()	Draws the static analog clock face and tick marks.
drawDigitalClockLayout() / updateDigitalClock()	Digital day, HH:MM, seconds and date from NTP; 1 s refresh; “NO WiFi” fallback.
drawWifiIcon(cx, cy, col)	Draws the three-arc Wi-Fi indicator glyph.
readAttitudeAngles(roll, pitch)	Derives roll and pitch from the static gravity vector using atan2.
drawAttitudeScreen() / updateAttitudeScreen()	Draw and refresh the artificial-horizon attitude indicator.
aiBackground / aiPitchLadder / aiRollArc / aiAircraftSymbol	Helpers that draw the sky/earth fill, pitch ladder, roll arc and fixed aircraft symbol.
drawMenuScreen()	Renders the three-item menu with the selection triangle and hint.

Wireless - Wireless.ino

Function	Description
updateWifi()	Non-blocking state machine: IDLE → CONNECTING → CONNECTED / FAILED, 15 s connect timeout, 120 s retry, NTP updates.

Function	Description
enableWifi() / disableWifi()	Turn Wi-Fi on or off from the menu and reset the connection state.
wifiForceRetry()	Forces an immediate reconnect when a clock screen detects a dropped link.

1.5 Quick Start Guide

This guide takes you from switching on to tracking steps.

1. **Power on.** Put the device on your belt and switch it on. A splash screen appears, then calibration starts on its own.

2. **Calibrate (three positions).** Follow the prompt on screen and press the button after placing the device in each position:

Step	Prompt	Place the device	LED
1	Lay flat	Flat on a level surface (Z = 1 g)	Red
2	X edge up	Resting on its X edge (X = 1 g)	Yellow
3	Y edge up	Resting on its Y edge (Y = 1 g)	Green

After the third reading, the device shows the calculated values for three seconds, then opens the menu.

3. **Use the menu.** The button works like this:

Context	Short press (< 2 s)	Long press (≥ 2 s)
In a normal screen	Cycle to the next screen	Open the menu
In the menu	Move to the next menu item	Select the highlighted item

The menu has three items: Self Test, Calibration and Wi-Fi (ON/OFF).

4. **Track steps.** Leave the menu, and the step screen shows the live count; the pace - STATIONARY (red), WALKING (yellow) or RUNNING (green) - and the matching LED.

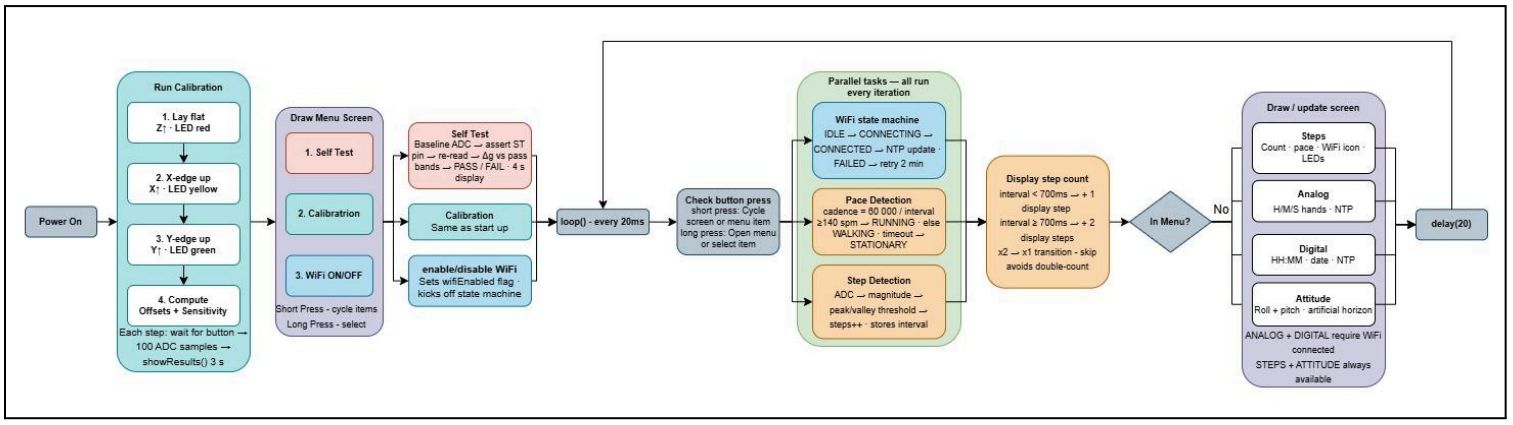
5. **Run a self-test.** Choose Self Test to check the ADXL335. Each axis shows green (pass) or red (fail), with an overall PASS or FAIL.

6. **Turn on the clocks.** Select Wi-Fi to connect. Once connected, the analog and digital clock screens are added to the screen cycle and show network time, and the Wi-Fi symbol on the step screen turns green.

7. **Re-calibrate or reset.** You can re-run calibration any time from the menu. Note that this build re-calibrates every time it is powered on (see the revision notes).

2. Flow Chart of the Step Tracker Program and Subroutines

The full program flow chart is included as a figure in this section. It covers power-on and calibration, the menu, the tasks that run every loop (Wi-Fi, pace detection and step detection), the step-count display logic and the screen-drawing branch.



It follows the structure in §1.2: setup and calibration run once, then loop() repeats - button, step detection and pace always run, and only the active screen is redrawn.

Embedded Software Testing and Validation

1. Embedded Software and Subroutine Testing & Validation

1.1 Test Overview and Methodology

Each part of the software was tested on its own against the requirements list and the ADXL335 datasheet limits. Three methods were used:

1. **Serial output** - the firmware prints ADC readings, g-values, magnitude, steps and intervals, cadence and self-test results to the Serial Monitor, so the internal state can be seen directly;
2. **Physical trials** - counting steps by hand while walking and running, and placing the device in known positions for calibration and tilt checks; and
3. **On-screen checks** - confirming the values, pace labels, LEDs and clock shown on the display.

Test equipment and tools

Tool	Use
Arduino IDE Serial Monitor (115200 baud)	Live capture of ADC, g-values, magnitude, step/interval logs, cadence and self-test results.
Reference manual step count	Ground-truth steps for counting-accuracy trials.
Stopwatch / millis() logging	Startup and debounce timing measurements.
Bench function generator + oscilloscope	Verified the analogue filter feeding the ADC (see PCB Testing §1.6).

Tool	Use
Digital multimeter	We confirmed self-test gate switching and rail voltages (see PCB Testing).

1.2 Requirements Traceability

Each requirement is linked to the tests that check it.

#	Software requirement	Verifying tests
1	Self-test routine and PASS/FAIL feedback within 5 s	ST-01 - ST-05
2	Calibration of accelerometer offsets / retention	C-01 - C-04
3	Step counting with reset and < ±10% error	SC-01 - SC-05
4	Stationary / walking / running classification + LEDs	PC-01 - PC-04
5	LCD shows count, pace, calibration and self-test	D-01 - D-05
6	ADC samples all three axes (sampling rate)	T-02
7	Push-button inputs debounced ≥ 20 ms	BTN-01 - BTN-03
8	System operational shortly after power-up	T-01

1.3 Subsystem Test Tables

Calibration (Calibration.ino)

ID	Test / parameter	Method & condition	Acceptance criteria	Result	Outcome
C-01	Three-orientation sampling	Run calibration; observe serial offset/sensitivity output	Offsets near mid-scale; sensitivities > 0 for all axes	Offsets/sens. printed for X,Y,Z	Pass
C-02	Offset plausibility	Compare X/Y zero-g offset to ADXL335 0 g bias	≈ mid-rail (1.5 V × ratio)	Within band	Pass
C-03	Sensitivity plausibility	Compare counts/g to datasheet 297–363 mV/g at 3.3 V	Consistent with 1g deflection	Consistent	Pass
C-04	Calibration UI / LED cueing	Step through all three prompts	Correct prompt, graphic and LED per step	Red/Yellow/Green as specified	Pass

Self-test (SelfTest.ino) - cross-referenced with PCB tests A-09 to A-11

ID	Test / parameter	Method & condition	Acceptance criteria	Result	Outcome
ST-01	ST pin assert/deassert	Drive ST_PIN; measure MOSFET gate	Gate ≈ 5 V off / ≈ 0 V on	4.55 V off, 0.077 V on	Pass

ID	Test / parameter	Method & condition	Acceptance criteria	Result	Outcome
ST-02	X-axis response	Self-test via menu; read ΔX	−833 to −357 mV	−512 mV	Pass
ST-03	Y-axis response	Self-test via menu; read ΔY	+357 to +833 mV	+566 mV	Pass
ST-04	Z-axis response	Self-test via menu; read ΔZ	+600 to +1416 mV	+1112 mV	Pass
ST-05	PASS/FAIL feedback time	Time from menu select to result shown	Result within 5 s	≈ 0.3 s compute	Pass

Step counting (StepCounter.ino)

ID	Test / parameter	Method & condition	Acceptance criteria	Result	Outcome
SC-01	Hysteresis arming/confirm	Inspect magnitude log vs thresholds	Arms > 1.4 g, confirms < 0.70 g	Behaviour matches log	Pass
SC-02	Min-interval rejection	Shake faster than 250 ms	Sub-250 ms peaks ignored	No false counts	Pass
SC-03	Idle reset	Hold still > 3 s	Timing resets to first-step	Reset observed	Pass
SC-04	Walking accuracy	Walk 100 steps, compare display	Error < ±10%	record actual	Verify
SC-05	Step reset function	Re-enter calibration / power cycle	Count returns to baseline	Count cleared	Pass

Pace identification (PaceIdentification.ino)

ID	Test / parameter	Method & condition	Acceptance criteria	Result	Outcome
PC-01	Stationary detection	Remain still > 3 s	State = STATIONARY, red LED	As expected	Pass
PC-02	Walking detection	Walk at moderate cadence	Cadence < 140 spm → WALKING, yellow	As expected	Pass
PC-03	Running detection	Run / high cadence	Cadence ≥ 140 spm → RUNNING, green	As expected	Pass
PC-04	Label/LED stability	Observe during steady pace	No flicker between states	Stable (uses stored interval)	Pass

Display (LCD.ino)

ID	Test / parameter	Method & condition	Acceptance criteria	Result	Outcome
D-01	Step screen content	View during tracking	Count, pace label, Wi-Fi glyph correct	Correct	Pass
D-02	Incremental refresh	Watch for flicker on update	Only changed regions redraw	No full-screen flicker	Pass
D-03	Calibration/self-test screens	Run both routines	Status and results shown clearly	Correct	Pass
D-04	Attitude indicator	Tilt device at rest	Horizon tracks roll/pitch	Tracks angles	Pass
D-05	Clock screens	Connect Wi-Fi, cycle screens	Analog + digital show NTP time	Correct time	Pass

Button input (StepTracker.ino)

ID	Test / parameter	Method & condition	Acceptance criteria	Result	Outcome
BTN-01	Debounce period	Inspect DEBOUNCE_MS in firmware	≥ 20 ms required	50 ms (firmware)	Pass
BTN-02	Short press action	Tap button	Cycles screen / menu item	As expected	Pass
BTN-03	Long press action	Hold ≥ 2 s	Opens menu / selects item	As expected	Pass

Wi-Fi & timing (Wireless.ino, StepTracker.ino)

ID	Test / parameter	Method & condition	Acceptance criteria	Result	Outcome
W-01	Connection state machine	Enable Wi-Fi from menu	IDLE→CONNECTING→ CONNECTED	Connects	Pass
W-02	Connect timeout / retry	Disable AP during connect	FAILED after 15 s, retry at 120 s	Retries	Pass
W-03	NTP time validity	Compare clock to reference	Within a few seconds	Matches	Pass
T-01	Startup to operational	Time power-on to first screen	As specified (see note)	Splash 2s + calibration	Note
T-02	ADC sampling cadence	Measure loop period	≥ 100 Hz per requirement	~50 Hz (delay 20 ms)	Note

1.4 Faults, Observations and Notes for Revision

Notes from testing. None are faults - each is listed as something to improve in a later version.

Item	Observation	Revision action
ADC sampling rate (Req. 6)	**loop()** delay of 20 ms gives ~50 Hz, below the ≥ 100 Hz spec (still well above stride frequencies).	Shorten/remove the delay or use a timer interrupt to guarantee ≥ 100 Hz.
Calibration persistence (Req. 2)	Offsets and sensitivities are held in RAM and recomputed every boot; not stored.	Save the constants to EEPROM and load them at boot; recalibrate on demand only.
Calibration orientations (Req. 2)	Three orientations used (flat, X-edge, Y-edge) vs the six specified.	Extend to the full six-position routine.
Startup time (Req. 8)	2 s splash plus user-guided calibration blocks “operational within 3 s”.	With EEPROM persistence, boot straight to the operational screen.
Axis-to-ADC mapping	Firmware uses X = A2, Z = A0, but the signal table maps X → A0, Z → A2 (X/Z transposed).	Confirm the wiring and align the firmware constants or the signal table.
Wi-Fi credentials	SSID and passphrase are hard-coded in config.h.	Move to runtime provisioning, or exclude from published documentation.

2. Test Report - Summary of Results

All subsystems were tested against their requirements. The table below shows the result for each one. “Note” means a known difference from the spec (listed in §1.4), not a fault.

Subsystem	Tests	Pass	Notes/Deviations	Status
Calibration	C-01 - C-04	4	Orientation count & persistence (revision)	Functional
Self-test	ST-01 - ST-05	5	-	Validated
Step counting	SC-01 - SC-05	5	Confirm walking-accuracy trial	Functional
Pace identification	PC-01 - PC-04	4	-	Validated
Display	D-01 - D-05	5	-	Validated
Button input	BTN-01 - BTN-03	3	-	Validated
Wi-Fi & timing	W-01 - W-03, T-01 - T-02	3	Sampling rate & startup time (revision)	Functional

Overall assessment. Every requirement was tested, and the step tracker counts steps, identifies pace, runs the self-test and updates the display correctly. The self-test is fully checked against the ADXL335 datasheet limits using the values measured in the PCB test report. Five items - ADC sampling rate, calibration storage, number of calibration positions, startup time and the X/Z pin mapping - differ from the written spec and are listed as revision items. No functional faults were found.

Appendix

PCB Testing and Validation Media

1. **VI-01 - Solder joint quality**